Online Appendix to "Intention-Based Reciprocity and Signalling of Intentions"

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A Theoretical Appendix

A.1 A model with symmetric preferences (see 2.2.2 in the main text)

In this subsection, I extend the model by also assuming social preferences for A. Similarly to B, assume that A's preferences are given by

$$u_A(\sigma,\, \sigma_B^*) = \mathbb{E}_{A,\sigma_B^*}\left[m_A(\sigma) + [\alpha_A + 1_{\{Meet\}}\theta_A]m_B(\sigma)\right]$$

where σ_B^* denotes A's prior belief that B will choose $Meet$, $\alpha_A \geq 0$ is A's altruistic preference parameter and $\theta_A \geq 0$ measures A's sensitivity to B's intentions. In this model, θ_A specifically measures the extent to which A cares about rewarding B for choosing the kind action $Meet$.[1]

As in the main text, we assume that B has preferences given by

$$u_B(\sigma,\, \sigma_A^*) = \mathbb{E}_B\left[m_B(\sigma) + [\alpha_B + 1_{\{In\}}\theta_B\mu_B(\sigma_A^*)]m_A(\sigma)\right]$$

where α_B, $\theta_B \geq 0$ might differ from α_A and θ_A.

Since B's preferences are unchanged compared to the main text, his optimal strategy remains the same. So to solve the model, consider A's choice. If Out is realized, A's payoff is $5(1+\alpha_A)$. If In is realized, A's expected utility is equal to $\sigma_B^*\left[10 + 10(\alpha_A + \theta_A)\right] + (1-\sigma_B^*)\left[2 + 14\alpha_A\right]$. Therefore, A chooses In if and only if

$$\sigma_B^*\left[10 + 10(\alpha_A + \theta_A)\right] + (1-\sigma_B^*)\left[2 + 14\alpha_A\right] \geq 5(1+\alpha_A)$$

$$\Leftrightarrow \left[8 - 4\alpha_A + 10\theta_A\right]\sigma_B^* \geq 3 - 9\alpha_A$$

To solve the model, we focus on the case where $\alpha_A < 2 + \frac{5}{2}\theta_A$ (so that $8 - 4\alpha_A + 10\theta_A > 0$).[2]

1. If $\alpha_A \geq \frac{1}{3}$, then A chooses In irrespective of her beliefs σ_B^*.
2. If $\alpha_A < \frac{1}{3}$, then A chooses In if and only if $\sigma_B^* \geq \bar{\sigma}(\alpha_A, \theta_A) := \frac{3-9\alpha_A}{8-4\alpha_A+10\theta_A} \in (0,1)$.

[1] One could also allow A to be negatively affected by B's choice to $Take$ by modelling A's preferences as

$$u_A(\sigma,\, \sigma_B^*) = \mathbb{E}_{A,\sigma_B^*}\left[m_A(\sigma) + [\alpha_A + \theta_A(1_{\{Meet\}} - 1_{\{Take\}})]m_B(\sigma)\right]$$

which would capture the idea of "betrayal aversion" (Bohnet et al. 2008).

[2] If $\alpha_A > 2 + \frac{5}{2}\theta_A$, it is easy to verify that A chooses In irrespective of her beliefs σ_B^*. Indeed, we have in this case $8 - 4\alpha_A + 10\theta_A < 0$ and therefore $3 - 9\alpha_A < 0$ ($\Leftrightarrow \alpha_A > \frac{1}{3}$). As a result, A chooses In iff $\sigma_B^* \leq \frac{3-9\alpha_A}{\mathbf{8-4\alpha_A+10\theta_A}}$, a ratio which is strictly above 1.

We have therefore the following set of equilibria:

CASE 1: $\alpha_B \geq \frac{1}{2}$
B chooses $Meet$. Thus $\sigma_B^* = \sigma_B = 1$ and A chooses In for any α_A, $\theta_A > 0$ so that $\sigma_A^* = \sigma_A = 1$. .

CASE 2: $\alpha_A \geq \frac{1}{3}$ and $\alpha_B + \theta_B < \frac{1}{2}$
A chooses In irrespective of B's action so that $\sigma_A^* = \sigma_A = 1$.
B chooses $Take$ (regardless of A's choice) and therefore $\sigma_B^* = \sigma_B = 0$.

CASE 3: $\alpha_A \geq \frac{1}{3}$ and $\alpha_B + \theta_B \geq \frac{1}{2}$
A chooses In irrespective of B's action so that $\sigma_A^* = \sigma_A = 1$.
B chooses $Meet$ since $\sigma_A^* > \bar{\sigma}(p, \alpha_B, \theta_B)$ and therefore $\sigma_B^* = \sigma_B = 1$.[3]

CASE 4: $\alpha_A < \frac{1}{3}$ and $\alpha_B + \theta_B < \frac{1}{2}$
B chooses $Take$ irrespective of A's action so that $\sigma_B^* = \sigma_B = 0$.
A chooses Out since $\sigma_B^* = 0$ and therefore $\sigma_A^* = \sigma_A = 0$.

The last case corresponds to the case where intentions matter for B:

CASE 5: $\alpha_A < \frac{1}{3}$, $\alpha_B + \theta_B \geq \frac{1}{2}$ and $\alpha_B < \frac{1}{2}$

1. When $p \in [0, \Lambda(\alpha_B, \theta_B))$, B chooses $Take$ irrespective of σ_A^* so that $\sigma_B^* = \sigma_B = 0$ and A chooses Out ($\sigma_A^* = \sigma_A = 0$).[4]

2. When $p \in [\Lambda(\alpha_B, \theta_B), 1)$, there are multiple equilibria:[5]

   (a) $\sigma_A^* = \sigma_A = 0$ and $\sigma_B^* = \sigma_B = 0$ is an equilibrium of self-fulfilling mistrust. It is the only equilibrium such that $\sigma_A^* < \bar{\sigma}(p, \alpha_B, \theta_B)$.

   (b) $\sigma_A^* = \sigma_A = 1$ and $\sigma_B^* = \sigma_B = 1$ is an equilibrium of self-fulfilling trust. It is the only equilibrium such that $\sigma_A^* > \bar{\sigma}(p, \alpha_B, \theta_B)$.

   (c) In addition, there is a mixed strategy equilibrium where $\sigma_A^* = \sigma_A = \bar{\sigma}(p, \alpha_B, \theta_B)$ and $\sigma_B^* = \sigma_B = \bar{\sigma}(\alpha_A, \theta_A)$.

3. When $p = 1$, there is a unique equilibrium where B chooses $Meet$ and A chooses In.

[3] Remember that $\bar{\sigma}(p, \alpha_B, \theta_B)$ corresponds to the threshold belief defined on page 9.
[4] For a definition of $\Lambda(\alpha_B, \theta_B)$, see page 9 of the main text.
[5] In the knife-edge case where $\alpha_B + \theta_B = \frac{1}{2}$, B chooses $Take$ for all $p \in [\Lambda(\alpha_B, \theta_B), 1)$.

A.2 Ex ante altruism and warm glow (see discussion of Section 5.1)

The discussion of Section 5.1 suggests that the relationship between prosocial behavior and the value of p might be affected by B's chances to determine the final outcome, since B had to make a decision only *assuming* that In was realized. To account for the use of the strategy method, I extend the model of Section 2 by relaxing the assumption that B observes whether In was realized before making a decision. Denote B's subjective belief in the event "In was realized" by $q(\sigma_A^*)$, that is, $q(\sigma_A^*) := p\sigma_A^* + \frac{1}{2}(1-p)$, and suppose that his preferences are given by:

$$U_B(\sigma_A^*, \sigma_B) = \mathbb{E}_B\left[m_B(\sigma) + [\alpha + 1_{\{In\}}\theta\mu_B(\sigma_A^*)]m_A(\sigma)\right] + \Phi(\sigma_B)$$

Now B's decision depends on two new components. First, B's decision depends on the probability that In is realized: B perceives that he only determines material payoffs with probability $q(\sigma_A^*)$, while with probability $1 - q(\sigma_A^*)$, A and B receive 5 dollars irrespective of B's decision. Second, B's preferences have a warm glow component $\Phi(\sigma_B)$, meaning that B cares about *appearing* altruistic (to himself or the experimenter), irrespective of whether his decision matters ex post. As such, $\Phi(.)$ only depends on the strategy chosen by B and is orthogonal to $q(\sigma_A^*)$. Assume that $\Phi(1) = \phi \geq 0$ and $\Phi(0) = 0$ i.e. B gets warm glow utility ϕ from choosing $Meet$ and 0 otherwise.

As in the main text, assume that players only adopt pure strategies and choose the pro-social action (In for A and $Meet$ for B) whenever indifferent. Therefore B chooses $Meet$ over $Take$ if and only if

$$U_B(\sigma_A^*, 1) \geq U_B(\sigma_A^*, 0)$$

$$\begin{aligned} \Leftrightarrow\ & 5(1+\alpha)[1 - q(\sigma_A^*)] + q(\sigma_A^*)[10 + 10(\alpha + \theta\mu_B(\sigma_A^*))] + \phi \\ \geq\ & 5(1+\alpha)[1 - q(\sigma_A^*)] + q(\sigma_A^*)[14 + 2(\alpha + \theta\mu_B(\sigma_A^*))] \end{aligned}$$

$$\Leftrightarrow q(\sigma_A^*)\left[8(\alpha + \theta\mu_B(\sigma_A^*)) - 4\right] + \phi \geq 0 \tag{1}$$

First, notice that when $q(\sigma_A^*) = 0$, B's decision does not affect the final outcome.[6] In this case, B chooses $Meet$ since $\phi \geq 0$. So suppose that $q(\sigma_A^*) > 0$. Then (1) can be rewritten as:

$$\mu_B(\sigma_A^*) \geq \frac{1-2\alpha}{2\theta} - \frac{\phi}{8\theta q(\sigma_A^*)} \tag{2}$$

Since comparative statics for the case where $\phi = 0$ were already derived in the main text, assume $\phi > 0$. From condition (2), it is easy to see that if $\alpha \geq \frac{1}{2}$ or $\phi \geq 4(1-2\alpha)$, altruism (pure or impure) overwhelms any other consideration so B chooses $Meet$ irrespective of his belief σ_A^* and the value of p. So assume for the rest of the analysis that $\alpha < \frac{1}{2}$ and $\phi < 4(1-2\alpha)$. Observing that $\mu_B(\sigma_A^*) = \frac{p\sigma_A^*}{q(\sigma_A^*)}$, (2) can be rewritten as:

[6] This can only happen if $p = 1$ and $\sigma_A^* = 0$ as is the case in (i) *EI-Out* and *COM-Out*, or (ii) NI and *No-COM* provided B believes $\sigma_A^* = 0$.

$$4p\,\sigma_A^*\,(\theta+\alpha-\frac{1}{2}) \geq (1-2\alpha)(1-p)-\frac{\phi}{2} \tag{3}$$

In the following, let p_ϕ be the value of p that solves $\phi = 2(1-2\alpha)(1-p)$ (that is, $p_\phi := 1-\frac{\phi}{2(1-2\alpha)}$).

Comparative statics with respect to p depend on the size of the warm glow effect:

1. **Small warm glow effect:** $\phi \in (0, 2(1-2\alpha))$

   - When $\theta+\alpha=\frac{1}{2}$, B chooses $Meet$ iff $p \geq p_\phi$, by condition (3).
   - When $\theta+\alpha>\frac{1}{2}$, (3) can be rewritten as:

$$\sigma_A^* \geq \bar{\sigma}(p,\alpha,\theta,\phi) \;\text{ where }\; \bar{\sigma}(p,\alpha,\theta,\phi) := \frac{(\frac{1}{2}-\alpha)(1-p)-\frac{1}{4}\phi}{2p(\theta+\alpha-\frac{1}{2})} \tag{4}$$

   Notice that (i) $\bar{\sigma}(p,\alpha,\theta,\phi) \leq 0$ iff $p \geq p_\phi$; (ii) $\bar{\sigma}(p,\alpha,\theta,\phi) > 1$ iff $p < \Lambda(\alpha,\theta,\phi) := \frac{1-2\alpha-\frac{\phi}{2}}{4\theta-(1-2\alpha)}$.

   Therefore, by condition (4):

   - If $p < \Lambda(\alpha,\theta,\phi)$, B chooses $Take$ irrespective of his beliefs.
   - If $p \in [\Lambda(\alpha,\theta,\phi), p_\phi)$, B's propensity to $Meet$ depends on σ_A^* and is *increasing* in p since[7]

$$\frac{\partial\bar{\sigma}(p,\alpha,\theta,\phi)}{\partial p} = \frac{\phi-2(1-2\alpha)}{8p^2(\theta+\alpha-\frac{1}{2})} < 0$$

   - If $p \geq p_\phi$, B chooses $Meet$ irrespective of his beliefs.

   - When $\theta+\alpha<\frac{1}{2}$, (3) can be rewritten as:

$$\sigma_A^* \leq \tilde{\sigma}(p,\alpha,\theta,\phi) \;\text{ where }\; \tilde{\sigma}(p,\alpha,\theta,\phi) := \frac{\frac{1}{4}\phi-(\frac{1}{2}-\alpha)(1-p)}{2p\,(\frac{1}{2}-\theta-\alpha)} \tag{5}$$

   Here (i) $\tilde{\sigma}(p,\alpha,\theta,\phi) < 0$ iff $p < p_\phi$; (ii) $\tilde{\sigma}(p,\alpha,\theta,\phi) > 1$ iff $p < \tilde{\Lambda}(\alpha,\theta,\phi) := \frac{\frac{\phi}{2}-(1-2\alpha)}{1-2\alpha-4\theta}$.

   Therefore, by condition (5):

   - If $p < p_\phi$, B chooses $Take$ irrespective of his beliefs.
   - If $p \in [p_\phi, \tilde{\Lambda}(\alpha,\theta,\phi))$, B chooses $Meet$ irrespective of his beliefs.[8]
   - If $p \geq \tilde{\Lambda}(\alpha,\theta,\phi)$, B's propensity to $Meet$ depends on σ_A^* and is *increasing* in p since

$$\frac{\partial\tilde{\sigma}(p,\alpha,\theta,\phi)}{\partial p} > 0$$

[7] It is easy to verify that the interval $[\Lambda(\alpha,\theta,\phi), p_\phi)$ is not empty and $\Lambda(\alpha,\theta,\phi) > 0$.

[8] Note that $\tilde{\Lambda}(\alpha,\theta,\phi) > 0$ if and only if $4\theta+2\alpha > 1$; under this condition, the interval $[p_\phi, \tilde{\Lambda}(\alpha,\theta,\phi))$ is non trivial.

2. **Large warm glow effect:** $\phi \in [2(1-2\alpha), 4(1-2\alpha))$

   - When $\theta + \alpha \geq \frac{1}{2}$, B chooses $Meet$ irrespective of his beliefs and the value of p. This easily follows from condition (3).

   - When $\theta + \alpha < \frac{1}{2}$, B chooses $Meet$ iff condition (5) holds:
     - if $p \in [0, \tilde{\Lambda}(\alpha, \theta, \phi))$, B chooses $Meet$ irrespective of σ_A^*.[9]
     - if $p \in [\tilde{\Lambda}(\alpha, \theta, \phi), 1]$, B's propensity to $Meet$ depends on σ_A^* and is *decreasing* in p since

$$\frac{\partial \tilde{\sigma}(p, \alpha, \theta, \phi)}{\partial p} < 0$$

Thus, there is a range in the parameter space where A's intentions matter little for B and the warm glow component is large enough (but not so large as to wipe out any other concern) to induce B to $Meet$ less often as p increases.

A.3 Comparative statics in other social preference models

This section analyzes comparative statics in the Noisy Trust Game $\Gamma_1(p, \Omega_A)$ for three of the most popular models in the literature on social preferences: the model of intention-based reciprocity of Dufwenberg and Kirchsteiger (2004), the model of simple guilt of Battigalli and Dufwenberg (2007) and the model of inequity aversion of Fehr and Schmidt (1999). To keep a certain symmetry with the model presented in the main text, I continue to assume that player A has selfish preferences and only modify the structure of B's preferences to match these three models.

In order to make meaningful comparisons, I assess the relative explanatory power of these models under the assumption of equilibrium behavior, which allows to make testable predictions without relying on beliefs that were not elicited in the experiment.[10] Furthermore, in the guilt aversion model, I study the set of sequential equilibria that survive a forward induction reasoning refinement formally introduced in Battigalli and Dufwenberg (2009). This additional refinement improves the explanatory power of the guilt aversion model, which would otherwise make weak predictions.[11] Table 1 (resp. 2) at the end of this section shows the percentage of A's (resp. B's) behavior that can be rationalized by each of the models analyzed in this section; the corresponding numbers for random play, selfish Nash and the model of Section 2.2 are also reported for reference.

[9] In this case, $\tilde{\Lambda}(\alpha, \theta, \phi) > 0$ if and only if $4\theta + 2\alpha < 1$.

[10] As in the main text, I continue to assume that individuals only play pure strategies ($\sigma_i \in \{0, 1\}$, $i \in \{\text{A,B}\}$) and choose the prosocial action (In for A and $Meet$ for B) whenever indifferent. Therefore, whenever studied, mixed strategies should be taken as coming from a statistical distribution of pure strategies played by individuals in role i who were drawn at random.

[11] This forward induction criterion does not affect the empirical performance of the other models.

A.3.1 Intention-Based Reciprocity à la Dufwenberg and Kirchsteiger (2004)

Following Dufwenberg and Kirchsteiger (2004) (henceforth DK), suppose first that player B is a reciprocal type with preferences given by

$$u_B(\sigma,\, \sigma_B^{**}) = m_B(\sigma) + \gamma \cdot \kappa_{BA}(\sigma_B) \cdot \hat{\kappa}_{BAB}(\sigma_A,\, \sigma_B^{**})$$

where $m_B(\sigma)$ is the Receiver's monetary payoff, $\gamma \geq 0$ captures his sensivity to reciprocity, κ_{BA} refers to the Receiver's kindness towards the Sender and $\hat{\kappa}_{BAB}$ corresponds to the Receiver's belief about the kindness of the Sender towards him. Perceived kindness is a function of B's belief about A's expectation that B will choose $Meet$, σ_B^{**}. With this functional form, a reciprocal Receiver reciprocates perceived (un-) kindness with (un-)kindness. The kindness of player A towards B is given by:

$$\kappa_{AB}(\sigma_A, \sigma_B^*) = \mathbb{E}_A\left[m_B|\sigma_B^*\right] - m_e(\sigma_B^*)$$

where $m_e(\sigma_B^*)$ is the equitable payoff:

$$m_e(\sigma_B^*) = \frac{\mathbb{E}_A\left[m_B|\sigma_B^*, \sigma_A = 1\right] + \mathbb{E}_A\left[m_B|\sigma_B^*, \sigma_A = 0\right]}{2}$$

In the Noisy Trust Game:
$\mathbb{E}_A\left[m_B|\sigma_B^*, \sigma_A = 1\right] = \left[p + \frac{1}{2}(1-p)\right]\left[10\sigma_B^* + 14(1-\sigma_B^*)\right] + \frac{1}{2}(1-p)\cdot 5$
$\mathbb{E}_A\left[m_B|\sigma_B^*, \sigma_A = 0\right] = \frac{1}{2}(1-p)\left[10\sigma_B^* + 14(1-\sigma_B^*)\right] + \left[p + \frac{1}{2}(1-p)\right]\cdot 5$
$m_e(\sigma_B^*) = \frac{10\sigma_B^* + 14(1-\sigma_B^*)}{2} = 9.5 - 2\sigma_B^* > 0\ \forall \sigma_B^* \in [0,1]$

Therefore the kindness of A towards B when choosing In ($\sigma_A = 1$) given beliefs σ_B^* is given by:
$\kappa_{AB}(1, \sigma_B^*) = \mathbb{E}_A\left[m_B|\sigma_B^*, \sigma_A = 1\right] - m_e(\sigma_B^*) = (4.5 - 2\sigma_B^*)p\ \geq 0\ \forall \sigma_B^*, p \in [0,1]$

while the kindness of A towards B when choosing Out ($\sigma_A = 0$) given beliefs σ_B^* is given by:
$\kappa_{AB}(0, \sigma_B^*) = \mathbb{E}_A\left[m_B|\sigma_B^*, \sigma_A = 0\right] - m_e(\sigma_B^*) = (2\sigma_B^* - 4.5)p\ \leq 0\ \forall \sigma_B^*, p \in [0,1]$

Then B's perception of A's kindness when she chose σ_A is given by

$$\hat{\kappa}_{BAB}(\sigma_A,\, \sigma_B^{**}) = \mathbb{E}_B\left[\kappa_{AB}(\sigma_A, \sigma_B^*)\right] = \mathbb{E}_B\left[m_B|\sigma_B^{**}\right] - m_e(\sigma_B^{**})$$

On the other hand, the kindness of B towards A is simply $\kappa_{BA} = 4$ $(= 10 - \frac{10+2}{2})$ if B chooses $Meet$ and $\kappa_{BA} = -4$ $(= 2 - \frac{10+2}{2})$ if B chooses $Take$.[12]

[12] In COM, the kindness of B towards A remains unchanged when A chooses to pay a fee: $\kappa_{BA} = 4$ $(= 9 - \frac{9+1}{2})$ if B chooses $Meet$ and $\kappa_{BA} = -4$ $(= 1 - \frac{9+1}{2})$ if B chooses $Take$.

I now analyze the comparative statics in each treatment. Let $\sigma_B^{**} := \mathbb{E}_B[\sigma_B^*]$ denote B's prior belief about A's expectation that B will choose $Meet$. In NI, $\Omega_A^1(S_A) = S_A$, that is, B learns nothing about A's action, so his posterior is equal to his prior σ_B^{**}. Let $\sigma_{B|In}^{**} := \mathbb{E}_B[\sigma_B^*|\sigma_A = 1]$ denote B's posterior belief about σ_B^* after A chose In in EI or COM and similarly define $\sigma_{B|Out}^{**} := \mathbb{E}_B[\sigma_B^*|\sigma_A = 0]$ as B's posterior belief if A chose Out. Finally, let $\sigma_{B|NC}^{**} := \mathbb{E}_B[\sigma_B^*|\text{No Communication}]$ denote B's posterior belief in COM if A did not choose to communicate her action.

In treatment NI, B's decision between $Meet$ and $Take$ depends on how likely he thinks A chose In when In was realized, because his perception of A's kindness depends on the action he thinks A chose. In particular, $\mathbb{E}_B[\hat{\kappa}_{BAB}(\sigma_A, \sigma_B^{**})] = \mu_B(\sigma_A^*)\kappa_{AB}(1, \sigma_B^{**}) + [1 - \mu_B(\sigma_A^*)]\,\kappa_{AB}(0, \sigma_B^{**})$ where $\mu_B(\sigma_A^*) = \frac{p\sigma_A^*}{p\sigma_A^* + \frac{1}{2}(1-p)}$. Therefore, B will choose $Meet$ if and only if

$$10 + 4\gamma \cdot \mathbb{E}_B[\hat{\kappa}_{BAB}(\sigma_A, \sigma_B^{**})] \geq 14 - 4\gamma \cdot \mathbb{E}_B[\hat{\kappa}_{BAB}(\sigma_A, \sigma_B^{**})]$$

$$\Leftrightarrow p\gamma \cdot (4.5 - 2\sigma_B^{**})[2\mu_B(\sigma_A^*) - 1] \geq \frac{1}{2} \quad (*)$$

As in the model developed in this paper, B's propensity to $Meet$ is weakly increasing in p since $\mu_B(\sigma_A^*)$ is strictly increasing in p, $\gamma \geq 0$ and $4.5 - 2\sigma_B^{**} > 0$ for all σ_B^{**}. This model makes however another prediction, which is not tested in this experiment: B's propensity to $Meet$ is decreasing in σ_B^{**}, that is, the more B thinks that A expects him to choose $Meet$, the less kind B perceives A's decision to go In.

A similar condition holds in $No\text{-}COM$ i.e. B will choose $Meet$ provided that

$$p\gamma \cdot (4.5 - 2\sigma_{B|NC}^{**})\left[2\mu_B(\sigma_{A|NC}^*) - 1\right] \geq \frac{1}{2}$$

where $\sigma_{A|NC}^* := \mathbb{E}_B[\sigma_A|\text{No Communication}]$.

In $EI\text{-}Out$ and $COM\text{-}Out$, B's belief about A's expectations is given by $\sigma_{B|Out}^{**}$ and B's perception of A's kindness is given by $\hat{\kappa}_{BAB}(0, \sigma_{B|Out}^{**})$. Using condition $(*)$ with $\sigma_{B|Out}^{**}$ replacing σ_B^{**} and $\sigma_A^* = 0$ replacing $\mu_B(\sigma_A^*)$, one can see that B chooses $Take$ for all values of p and $\sigma_{B|Out}^{**}$.

In $EI\text{-}In$ and $COM\text{-}In$, B's belief about A's expectations is given by $\sigma_{B|In}^{**}$ and B's perception of A's kindness is given by $\hat{\kappa}_{BAB}(1, \sigma_{B|In}^{**})$. Replacing σ_B^{**} with $\sigma_{B|In}^{**}$ and $\mu_B(\sigma_A^*)$ with $\sigma_A^* = 1$, condition $(*)$ becomes

$$p\gamma \cdot (4.5 - 2\sigma_{B|In}^{**}) \geq \frac{1}{2}$$

Therefore, as in NI and $No\text{-}COM$, B's propensity to $Meet$ is increasing in p and decreasing in B's

second-order belief. For $\gamma > 0$, the above expression is equivalent to

$$\sigma^{**}_{B|In} \leq \bar{\sigma}(p, \gamma)$$

where $\bar{\sigma}(p, \gamma) := \frac{9}{4} - \frac{1}{4p\gamma}$. Notice that $\bar{\sigma}(p, \gamma) < 0$ if and only if $p < \frac{1}{9\gamma}$ and $\bar{\sigma}(p, \gamma) \geq 1$ if and only if $p \geq \frac{1}{5\gamma}$. Therefore:

- For $p \in [0, \frac{1}{9\gamma})$, B chooses $Take$ irrespective of A's choice.
- For $p \in [\frac{1}{9\gamma}, \frac{1}{5\gamma})$, B chooses $Meet$ in $EI\text{-}In$ and $COM\text{-}In$ if and only if $\sigma^{**}_{B|In} \leq \bar{\sigma}(p, \gamma)$ and B chooses $Take$ in $EI\text{-}Out$ and $COM\text{-}Out$.
- For $p \geq \frac{1}{5\gamma}$, B chooses $Meet$ in $EI\text{-}In$ and $COM\text{-}In$ and chooses $Take$ in $EI\text{-}Out$ and $COM\text{-}Out$.

Therefore this model reproduces Prediction 1 in the main text: for all treatments, B's propensity to $Meet$ and therefore A's propensity to go In are weakly monotone increasing in p.

Comparing behavior across treatments, B's propensity to $Meet$ in NI ($No\text{-}COM$) should be weakly lower compared to $EI\text{-}In$ ($COM\text{-}In$) and weakly higher compared to $CI\text{-}Out$ ($COM\text{-}Out$).[13] These findings correspond to Prediction 2(a) and 3(a) in the main text.

As a result of Prediction 2(b), A's propensity to go In should be weakly higher in EI than in NI (Prediction 2(a)). This model also generates Prediction 3(b) under the assumption of equilibrium behavior: A's propensity to communicate In as a function of p will be either flat or inverted U-shaped. To see why the last statement holds, notice that A will never choose to pay if she believes that B will $Meet$ in $No\text{-}COM$. On the other hand, it is optimal for A to pay if

$$p\gamma \cdot (4.5 - 2\sigma^{**}_{B|In}) \geq \frac{1}{2} > p\gamma \cdot (4.5 - 2\sigma^{**}_{B|NC}) \left[2\mu_B(\sigma^{*}_{A|NC}) - 1\right]$$

Notice that since $\mu_B(\sigma^{*}_{A|NC})$ is increasing in p, the benefit of signalling is decreasing in p. In the limit when $p = 1$, the only possibility for this condition to be satisfied is that $\sigma^{**}_{B|In} < \sigma^{**}_{B|NC}$, since $\mu_B(\sigma^{*}_{A|NC}) = 1$. But in equilibrium it must be that $\sigma^{**} = \sigma^{*} = \sigma$ and $\sigma^{*}_{B|In} < \sigma^{*}_{B|NC}$ cannot be an equilibrium condition, for A would then choose not to pay.

To complete the analysis, I now describe the set of equilibria in each treatment. I restrict attention to equilibria in pure strategy, although mixed strategy equilibria also exist. Starting with NI, it can be verified that the equilibrium of self-fulfilling trust in which $\sigma_B = \sigma^{*}_B = \sigma^{**}_B = 1$ and

[13] Comparing conditions for NI and $EI\text{-}In$, it is easy to verify that $p\gamma \cdot (4.5 - 2\sigma^{**}_{B|In}) > p\gamma \cdot (4.5 - 2\sigma^{**}_{B}) \left[2\mu_B(\sigma^{*}_{A}) - 1\right]$ for all values of $\mu_B(\sigma^{*}_{A})$, σ^{**}_{B} and $\sigma^{**}_{B|In}$. Therefore B's propensity to $Meet$ is always weakly higher in $EI\text{-}In$ compared to NI. A similar statement holds comparing $COM\text{-}In$ to $No\text{-}COM$.

$\sigma_A = \sigma_A^* = \sigma_A^{**} = 1$ can only be supported if

$$p\gamma(4.5 - 2 \times 1)[2\mu_B(1) - 1] \geq \frac{1}{2}$$

$$\Leftrightarrow \quad 5p\gamma\left(\frac{3p-1}{1+p}\right) \geq 1$$

$$\Leftrightarrow \quad p \geq \max\left\{\frac{1}{3}, \zeta(\gamma)\right\} \; = \zeta(\gamma)$$

where $\zeta(\gamma) = \frac{1+5\gamma+\left[(1+5\gamma)^2+60\gamma\right]^{\frac{1}{2}}}{30\gamma}$. Since $\zeta(\gamma) \leq 1 \Leftrightarrow \gamma \geq \frac{1}{5}$, we have the following proposition:

Proposition DK-1 (treatment NI)

1. *When $\gamma \in [0, \frac{1}{5})$, there is a unique equilibrium such that $\sigma_A = \sigma_A^* = \sigma_A^{**} = 0$ and $\sigma_B = \sigma_B^* = \sigma_B^{**} = 0$. In this equilibrium of self-fulfilling mistrust, the Sender's decision not to invest is justified by the Receiver's decision to take if he invests.*

2. *When $\gamma \in [\frac{1}{5}, \infty)$, there exists in addition to the previous equilibrium of self-fulfilling mistrust a pure strategy equilibrium with self-fulfilling trust ($\sigma_A = \sigma_A^* = \sigma_A^{**} = 1$ and $\sigma_B = \sigma_B^* = \sigma_B^{**} = 1$) provided that $p \geq \zeta(\gamma)$.*

Turning to EI, B will $Take$ after $\sigma_A = 0$ in any equilibrium of the game. On the other hand, B will $Meet$ after $\sigma_A = 1$ provided that

$$p\gamma \cdot (4.5 - 2 \times 1) \geq \frac{1}{2} \;\Leftrightarrow\; p \geq \frac{1}{5\gamma}$$

which can only happen if $\frac{1}{5\gamma} \leq 1$ i.e. $\gamma \geq \frac{1}{5}$. To lighten notation, I omit in the remaining propositions the equilibrium consistency requirement $\sigma_{B|h} = \sigma_{B|h}^* = \sigma_{B|h}^{**}$ for all $h \in \mathcal{H}$.

Proposition DK-2 (treatment EI)

1. *When $\gamma \in [0, \frac{1}{5})$, there is a unique equilibrium of self-fulfilling mistrust such that $\sigma_A = 0$ and $\sigma_{B|h} = 0$ for all $h \in \{In, Out\}$.*

2. *When $\gamma \in [\frac{1}{5}, \infty)$, the equilibrium profile depends on the value of p:*

   (a) *If $p \in [0, \frac{1}{5\gamma})$, there is a unique equilibrium of self-fulfilling mistrust, where $\sigma_A = 0$ and $\sigma_{B|h} = 0$ for all $h \in \{In, Out\}$.*

   (b) *If $p \in [\frac{1}{5\gamma}, 1]$, there is a unique equilibrium of self-fulfilling trust, where $\sigma_A = 1$, $\sigma_{B|In} = 1$* and $\sigma_{B|Out} = 0$.

It can be verified that $\zeta(\gamma) \geq \frac{1}{5\gamma}$ for all $\gamma \geq \frac{1}{5}$. Therefore, self-fulfilling trust is more likely to occur in EI than in NI.

In COM, again there is a unique equilibrium when $\gamma < \frac{1}{5}$ such that B chooses $Take$ after any history and A chooses (Out, NC). As we learned from the analysis of EI, B will also choose $Take$ when $\gamma \geq \frac{1}{5}$ if the value of p is too low ($p < \frac{1}{5\gamma}$). On the other hand, when $\gamma \geq \frac{1}{5}$ and $p \geq \frac{1}{5\gamma}$, B will choose $Meet$ if A's strategy is (In, C). Therefore, whether A chooses to pay to communicate In depends on B's behavior after history $h = NC$. Similarly to treatment NI, B will $Meet$ (and therefore A will go In without communication) provided that $p \geq \zeta(\gamma)$; otherwise, B will $Take$ after $h = NC$. Therefore we have the following proposition:

Proposition DK-3 (treatment COM)

1. *When $\gamma \in [0, \frac{1}{5})$, there is a unique equilibrium of self-fulfilling mistrust such that* A chooses *(Out, NC) and B chooses $Take$ after any history, i.e. $\sigma_{B|h} = 0$ for all $h \in \{In, Out, NC\}$.*

2. *When $\gamma \in [\frac{1}{5}, \infty)$, the equilibrium profile depends on the value of p:*

   (a) *If $p \in [0, \frac{1}{5\gamma})$, there is a unique equilibrium of self-fulfilling mistrust, where A chooses (Out, NC) and $\sigma_{B|h} = 0$ for all $h \in \{In, Out, NC\}$.*

   (b) *If $p \in [\frac{1}{5\gamma}, \zeta(\gamma))$, there is a unique equilibrium of self-fulfilling trust with signalling, where* A chooses (In, C), $\sigma_{B|In} = 1$ and $\sigma_{B|h} = 0$ *for $h \in \{Out, NC\}$.*

   (c) If $p \in [\zeta(\gamma), 1]$, *there are only equilibria of self-fulfilling trust with no signalling, where* A chooses (In, NC), $\sigma_{B|NC} = 1$, $\sigma_{B|Out} = 0$ *and* $\sigma_{B|In} \in [0, 1]$.

We finally notice that this model can account for the *always Out* (*Take*) and the *monotone increasing* types. It cannot account for the types who choose *always In* (*Meet*). To see the intuition, notice that when $p = 0$, A has no way to be kind to B and therefore B will only seek to maximize his material payoffs.

A.3.2 Guilt aversion à la Battigalli and Dufwenberg (2007)

Now assume that the preferences of B are of the form

$$u_B(\sigma, \sigma_B^{**}) = m_B(\sigma) + \gamma \sigma_B^{**} m_A(\sigma)$$

where $\gamma > 0$ measures B's "trust responsiveness", that is, the extent to which B cares about meeting A's expectations. One can think of trust responsiveness as a moral obligation to fulfill trust or as an aversion to letting down an A player who trusted. This specification is formally equivalent to

the model of linear guilt of Battigalli and Dufwenberg (2007) (henceforth BD). As in the model of intention-based reciprocity of DK 2004, a key feature of this model is B's second order belief σ_B^{**}. However, in contrast with DK 2004, σ_B^{**} enters B's utility positively: the more B thinks that A expects him to choose $Meet$, the more likely he is to select $Meet$.

Formally, let $\sigma_B^{**} := \mathbb{E}_B[\sigma_B^*] = \int_0^1 \sigma_B^* dF(\sigma_B^*)$ be B's prior belief about A's belief σ_B^*. As in A.3.1, let $\sigma_{B|In}^{**} := \mathbb{E}_B[\sigma_B^*|\sigma_A = 1]$ denote B's posterior belief about σ_B^* after A chose In in NI or COM and similarly define $\sigma_{B|Out}^{**} := \mathbb{E}_B[\sigma_B^*|\sigma_A = 0]$ as B's posterior belief after A chose Out. Finally, let $\sigma_{B|NC}^{**} := \mathbb{E}_B[\sigma_B^*|\text{No Communication}]$ denote B's posterior belief in COM if A did not choose to communicate her action. To make comparative statics predictions, I make the following assumption:

Assumption 1: $\mathbb{E}_B[\sigma_B^*|\sigma_B^* \geq k(p)]$ is strictly increasing in k[14] and differentiable with respect to p.

Note that in any treatment and after any history $h \in \mathcal{H}$ (including when A communicated her action in COM i.e. $1_{\{C\}} = 1$), B will choose $Meet$ if and only if

$$10 + (10 - 1_{\{C\}})\gamma\sigma_{B|h}^{**} \geq 14 + (2 - 1_{\{C\}})\gamma\sigma_{B|h}^{**}$$

$$\Leftrightarrow \ \sigma_{B|h}^{**} \geq \frac{1}{2\gamma} \qquad (CB)$$

In NI, A will choose In if and only if

$$[p + \frac{1}{2}(1-p)][10\sigma_B^* + 2(1-\sigma_B^*)] + \frac{1}{2}(1-p)\cdot 5 \geq \frac{1}{2}(1-p)[10\sigma_B^* + 2(1-\sigma_B^*)] + [p + \frac{1}{2}(1-p)]\cdot 5$$

$$\Leftrightarrow \ \sigma_B^* \geq \tfrac{3}{8} \qquad (CA\text{-}1)$$

In CI, A will choose In if and only if

$$[p+\frac{1}{2}(1-p)][10\sigma_{B|In}^* + 2(1-\sigma_{B|In}^*)] + \frac{1}{2}(1-p)\cdot 5 \geq \frac{1}{2}(1-p)[10\sigma_{B|Out}^* + 2(1-\sigma_{B|Out}^*)] + [p+\frac{1}{2}(1-p)]\cdot 5$$

$$\Leftrightarrow \ \sigma_{B|In}^* \geq \frac{3p + 4(1-p)\sigma_{B|Out}^*}{4(1+p)} \qquad (CA\text{-}2)$$

Similar computations (see Section B.3 of this OA) show that in COM, A will choose (In, C) over any other strategy in $\{(In, NC), (Out, NC), (Out, C)\}$ if and only if

$$\sigma_{B|In}^* \geq \max\left\{\sigma_{B|NC}^* + \frac{1}{8}, \frac{7p+1}{8(1+p)} + \frac{1-p}{1+p}\sigma_{B|NC}^*, \frac{p}{1+p} + \frac{1-p}{1+p}\sigma_{B|Out}^*\right\} \qquad (CA\text{-}3)$$

while A will choose (Out, C) provided

[14] A sufficient condition is to assume that F is a distribution function with full support.

$$\sigma^*_{B|Out} \geq \max\left\{\sigma^*_{B|NC} + \frac{1}{8}, \frac{1-7p}{8(1-p)} + \frac{1-p}{1+p}\sigma^*_{B|NC}, \frac{1+p}{1-p}\sigma^*_{B|In} - \frac{p}{1-p}\right\} \qquad (CA\text{-}4)$$

and finally A will prefer (In, NC) over (Out, NC) provided $\sigma^*_{B|NC} \geq \frac{3}{8}$.

Notice that when $\gamma < \frac{1}{2}$, condition (CB) is never satisfied, meaning B chooses $Take$ regardless of his beliefs about A's expectations. Therefore, the only equilibrium is $\sigma^{**}_B = \sigma^*_B = \sigma_B = 0$ and $\sigma_A = 0$. When $\gamma \geq \frac{1}{2}$, this psychological game has a multiplicity of equilibria in all treatments, depending on players' beliefs. However, for large enough values of γ, the refinement of forward induction imposes restrictions on the equilibrium set (see Battigalli and Dufwenberg 2009). For treatment NI, notice that by forward induction reasoning and condition CA-1, B must believe that $\sigma^{**}_B \geq \frac{3}{8}$. Therefore, condition (CB) will be strictly satisfied whenever $\gamma > \frac{4}{3}$, leading to the unique equilibrium $\sigma^{**}_B = \sigma^*_B = \sigma_B = 1$ and $\sigma_A = 1$. This is summarized in the following proposition:

Proposition BD-1 (treatment NI)

1. *For $\gamma \in [0, \frac{1}{2})$, there is a unique equilibrium such that $\sigma_A = 0$ and $\sigma_B = \sigma^*_B = \sigma^{**}_B = 0$. In this equilibrium of self-fulfilling mistrust, the Sender's decision not to invest is justified by the Receiver's decision to take if he invests.*

2. *For $\gamma \in [\frac{1}{2}, \frac{4}{3}]$, there are multiple equilibria. In addition to the previous equilibrium of self-fulfilling mistrust ($\sigma_A = 0$ and $\sigma_B = \sigma^*_B = \sigma^{**}_B = 0$), there is a pure strategy equilibrium with self-fulfilling trust ($\sigma_A = 1$ and $\sigma_B = \sigma^*_B = \sigma^{**}_B = 1$). Furthermore, for $\gamma \neq \frac{1}{2}$, there is a mixed-strategy equilibrium such that $\sigma_B = \sigma^*_B = \sigma^{**}_B = \frac{1}{2\gamma}$ and the Sender chooses In with positive probability.*

3. *For $\gamma \in (\frac{4}{3}, \infty)$, forward induction reasoning leads to a unique equilibrium of self-fulfilling trust ($\sigma_A = 1$ and $\sigma_B = \sigma^*_B = \sigma^{**}_B = 1$).*

Notice here that even under the restrictions of forward induction reasoning, the guilt aversion model makes no prediction as to the relationship between prosocial behavior (choice of In and $Meet$) and the value of p, since conditions (CB) and $(CA\text{-}1)$ do not depend on p.

In the other two treatments, the set of equilibria is larger since a strategy for B, $\sigma_B := \left(\sigma_{B|h}\right)_{h\in\mathcal{H}}$, now has to specify an action after any history $h \in \mathcal{H}$ where $\mathcal{H} = \{In, Out\}$ in CI and $\mathcal{H} = \{In, Out, NC\}$. To lighten notation, I omit in the following propositions the equilibrium consistency requirement $\sigma_{B|h} = \sigma^*_{B|h} = \sigma^{**}_{B|h}$ for all $h \in \mathcal{H}$.

In treatment EI, note that the RHS of condition CA-2 is increasing in p provided that $\sigma^*_{B|Out} < \frac{3}{4}$. Under forward induction reasoning and given Assumption 1, B's second-order belief $\sigma^{**}_{B|In} \geq \mathbb{E}_B\left[\sigma^*_B | \sigma^*_B \geq \frac{3p}{4(1+p)}\right]$ is therefore strictly increasing in p in any equilibrium in which $\sigma_{B|Out} < \frac{3}{4}$, a restriction which is consistent with the data. In particular, forward induction reasoning eliminates all equilibria except those of self-fulfilling trust for all values of γ such that $\frac{3p}{4(1+p)} > \frac{1}{2\gamma}$ i.e. $\gamma > \frac{2}{3}\left(1+\frac{1}{p}\right)$. The following proposition summarizes this finding:

Proposition BD-2 (treatment EI)

1. *For $\gamma \in [0, \frac{1}{2})$, there is a unique equilibrium of self-fulfilling mistrust, given by $\sigma_A = 0$ and $\sigma_{B|h} = 0$ for all $h \in \{In, Out\}$.*

2. *For $\gamma \in \left[\frac{1}{2}, \frac{2}{3}\left(1+\frac{1}{p}\right)\right]$, there are multiple equilibria. The set Σ of pure strategy equilibria $(\sigma_A, \sigma_{B|In}, \sigma_{B|Out})$ is given by $\Sigma = \{(0,0,0),(1,1,0),(1,1,1),(0,0,1)\}$. In addition, for $\gamma \neq \frac{1}{2}$, there is a set of mixed-strategy equilibria in which $\sigma_{B|h} = \frac{1}{2\gamma}$, $\sigma_{B|h'} \in \{0, \frac{1}{2\gamma}, 1\}$ and $\sigma_A > 0$ if and only if $u_A(1, (\sigma_{B|h})_{h\in\mathcal{H}}) \geq u_A(0, (\sigma_{B|h})_{h\in\mathcal{H}})$.*

3. *For $\gamma \in \left(\frac{2}{3}\left(1+\frac{1}{p}\right), \infty\right)$, forward induction reasoning eliminates all equilibria except those of self-fulfilling trust, where $\sigma_A = 1$, $\sigma_{B|In} = 1$ and $\sigma_{B|Out} \in [0,1]$.*

Notice that since $\frac{2}{3}\left(1+\frac{1}{p}\right)$ is decreasing in p, self-fulfilling trust is more likely to emerge as p increases. Furthermore, comparing predictions in NI and EI, notice that $\frac{2}{3}\left(1+\frac{1}{p}\right) > \frac{4}{3}$ for all $p \in [0,1)$ and $\frac{2}{3}\left(1+\frac{1}{p}\right) = \frac{4}{3}$ when $p = 1$.[15] Therefore under the restriction of forward induction reasoning, an equilibrium of self-fulfilling trust is more likely to emerge in NI than in EI. To see the intuition, notice that the RHS of CA-1 is higher than the RHS of CA-2 provided that $\sigma^*_{B|Out} < \frac{3}{8}$. By Assumption 1, it must be in this case that $\sigma^{**}_{B|In} < \sigma^{**}_B$, implying that forward induction reasoning imposes stronger restrictions on B's beliefs in NI than EI. Intuitively, if A expects to be punished conditional on choosing Out in EI, then choosing In is a less strong signal of trust than in the NI environment where B cannot observe A's action and acts according to his prior σ^*_B. Thus, when A chooses In to avoid being punished, B is less likely to feel guilty than when A chooses In, trusting that B will act benevolently under his prior.

Turning to COM, notice first that when $\sigma^*_{B|NC} \geq \frac{7}{8}$, A's optimal choice is (In, NC). Therefore, in any equilibrium of the game in which A chooses (In, C) (or (Out, C)) with positive probability, it must be that $\sigma_{B|NC} < \frac{7}{8}$. Notice that when $\sigma^*_{B|NC} < \frac{7}{8}$, the RHS in CA-3 is increasing in p, while the RHS in CA-4 is decreasing in p. Thus, under forward induction reasoning and Assumption 1,

[15] Since $p = 0$ implies that $\gamma = \infty$, no equilibrium can be eliminated under forward induction reasoning when A's action is not implemented.

B's second-order belief $\sigma^{**}_{B|In}$ ($\sigma^{**}_{B|Out}$) is strictly increasing (decreasing) in p in any equilibrium in which $\sigma_{B|NC} < \frac{7}{8}$. Furthermore, since $\max\left\{\frac{1}{8}, \frac{7p+1}{8(1+p)}, \frac{p}{1+p}\right\} = \frac{7p+1}{8(1+p)}$ for all $p \in [0,1]$ and observing that $\frac{7p+1}{8(1+p)} > \frac{1}{2\gamma} \Leftrightarrow \gamma > \frac{4(1+p)}{7p+1}$, forward induction reasoning eliminates all equilibria in which A chooses Out with positive probability for all values of $\gamma > \frac{4(1+p)}{7p+1}$. In the following proposition, I restrict attention to pure strategy equilibria, although mixed strategy equilibria also exist:

Proposition BD-3 (treatment COM)

1. *For $\gamma \in [0, \frac{1}{2})$, there is a unique equilibrium of self-fulfilling mistrust, where A chooses (Out, NC) and B chooses Take after any history i.e. $\sigma_{B|h} = 0$ for all $h \in \{In, Out, NC\}$.*

2. *For $\gamma \in \left[\frac{1}{2}, \frac{4(1+p)}{7p+1}\right]$, there are multiple equilibria:*

   (a) *Self-fulfilling mistrust with no signalling: A chooses (Out, NC) and $\sigma_{B|h} = 0$ for all $h \in \{In, Out, NC\}$.*

   (b) *Self-fulfilling trust with no signalling: A chooses (In, NC), $\sigma_{B|NC} = 1$ and $(\sigma_{B|In}, \sigma_{B|Out}) \in \{0,1\} \times \{0,1\}$.*

   (c) *Self-fulfilling trust with signalling: A chooses (In, C), $\sigma_{B|In} = 1$, $\sigma_{B|NC} = 0$ and $\sigma_{B|Out} \in \{0,1\}$.*

   (d) *Rewarded signalling of mistrust: A chooses (Out, C) and $(\sigma_{B|In}, \sigma_{B|Out}, \sigma_{B|NC}) = (0,1,0)$.*

3. *For $\gamma \in \left(\frac{4(1+p)}{7p+1}, \infty\right)$, forward induction eliminates all equilibria in which A chooses Out:*

   (a) If $\sigma^*_{B|NC} < \frac{7}{8}$, then A chooses (In, C), $\sigma_{B|In} = 1$, $\sigma_{B|NC} = 0$ *and* $\sigma_{B|Out} \in \{0,1\}$.

   (b) If $\sigma^*_{B|NC} \geq \frac{7}{8}$, then A chooses (In, NC), $\sigma_{B|NC} = 1$ *and* $(\sigma_{B|In}, \sigma_{B|Out}) \in \{0,1\} \times \{0,1\}$.

It is easy to see that $\frac{4(1+p)}{7p+1}$ is decreasing in p; therefore, under the restrictions of forward induction reasoning, B is more likely to $Meet$ and A is more likely to go In as p increases. However, the model makes no predictions as to the relative likelihood of (In, C) and (In, NC) as p increases.

We finally notice that this model can account for the *always Out* (*Take*), the *always In* (*Meet*) and the *monotone increasing* types in EI and COM but does not predict monotone increasing behavior with respect to p in NI. Furthermore, the model makes no predictions regarding B's behavior after knowing that A chose Out (EI-Out and COM-Out).

A.3.3 Inequity aversion à la Fehr and Schmidt (1999)

Assume that the Receiver's utility is given by

$$u_B(\sigma) = m_B(\sigma) - \alpha_B \cdot \max\{m_A(\sigma) - m_B(\sigma),\, 0\} - \beta_B \cdot \max\{m_B(\sigma) - m_A(\sigma),\, 0\}$$

where $\alpha_B \geq \beta_B$ and $\beta_B \in [0, 1)$. Parameter α_B (resp. β_B) captures the Receiver's disutility from being behind (resp. ahead).

Here I assume that player B cares about his expected material payoffs relative to A. Because subjects in role B do not observe Nature's move before making a choice, they might indeed reason in expectation.[16] Notice that if B chooses $Meet$, A and B receive the same expected payoffs irrespective of A's choice; in this case, B suffers no disutility from inequity. On the other hand, if he chooses $Take$ when In is realized, B suffers a disutility $-12\beta_B$ from being ahead. Therefore, assuming that A chooses In, B will choose $Meet$ over $Take$ if and only if

$$[p + \frac{1}{2}(1-p)] \cdot 10 + \frac{1}{2}(1-p) \cdot 5 \geq [p + \frac{1}{2}(1-p)] \cdot 14 + \frac{1}{2}(1-p) \cdot 5 - [p + \frac{1}{2}(1-p)] \cdot 12\beta_B$$

$$\Leftrightarrow \beta_B \geq \frac{1}{3}$$

Similarly if A chooses Out, B will choose $Meet$ over $Take$ if and only if

$$\frac{1}{2}(1-p) \cdot 10 + [p + \frac{1}{2}(1-p)] \cdot 5 \geq \frac{1}{2}(1-p) \cdot 14 + [p + \frac{1}{2}(1-p)] \cdot 5 - \frac{1}{2}(1-p) \cdot 12\beta_B$$

$$\Leftrightarrow \beta_B \geq \frac{1}{3}$$

Consequently B's propensity to choose $Meet$ is insensitive to p and to the information he possesses about A's action.[17] The reason is that outside payoffs are identical for A and B. On the other hand, with unequal outside payoffs (for instance $m_A = 5$ and $m_B = 0$), it can be verified that B's propensity to $Meet$ would be:[18]

- *increasing* in p in *EI-In* and *COM-In*.
- *decreasing* in p in *EI-Out* and *COM-Out*.
- strictly higher in *EI-In* (*COM-In*) compared to *EI-Out* (*COM-Out*) for all $p \in [0, 1]$.

[16] On the other hand, if B only cared about realized payoffs, it is easy to see that B's decision would be orthogonal to p regardless of the payoff structure.

[17] Notice that even when A pays to communicate her action, B will choose $Meet$ if and only if $\beta_B \geq \frac{1}{3}$. Here B suffers a cost of $-\beta_B$ $(= -\beta_B(10-9))$ when choosing $Meet$ and of $-13\beta_B$ $(= -\beta_B(14-1))$ when choosing $Take$, implying a difference of $-12\beta_B$, as in the other treatment conditions.

[18] Other comparative statics generally depend on the relative size of α_B and β_B as well as on B's belief σ_A^* ($\sigma_{A|NC}^*$) in the NI (*No-COM*) condition.

The equilibrium profiles for the payoff structure of the experiment are summarized below:

Proposition FS (all treatments)

1. *When* $\beta_B < \frac{1}{3}$, *there exists a unique equilibrium in pure strategies where* $\sigma_A = 0$ *and* $\sigma_B = 0$.
2. *When* $\beta_B \geq \frac{1}{3}$, *there exists a unique equilibrium in pure strategies where* $\sigma_A = 1$ *and* $\sigma_B = 1$.
3. *In addition, when* $\beta_B = \frac{1}{3}$, *there is a mixed strategy equilibrium such that* $\sigma_B = \frac{3}{8}$ *and A chooses In with positive probability.*

Therefore, this model can account for the *always Out* (*Take*) and the *always In* (*Meet*) types.

A.3.4 Comparison of the models

Here I assess the explanatory power of the models analyzed above relative to the model presented in the main text (Section 2.2). Table 1 (resp. 2) shows the percentage of A's (resp. B's) behavior that can be explained by each of the theories; as benchmarks, I also report the corresponding percentages for random play and selfish Nash behavior. The classification of subjects into types refers to the one presented in Section 5.1 (see Table 5 and 6).

Table 1: % of A's behavior explained by other models (see Table 5 in main text)

Model	Rationalizable patterns	*All*	*NI*	*EI*	*COM*
Random play	essentially *non monotone*	20.4	15.8	8.3	35.9
Selfish	*always Out*	20.4	26.3	19.4	15.4
Inequity aversion	*always In*, *always Out*	45.1	50.0	47.2	38.5
Sequential reciprocity	*always Out*, *monotone increasing*	43.4	47.4	47.2	35.9
Guilt aversion	*always In*, *always Out* in *All* *monotone increasing* in *EI* and *COM*	61.1	50.0	75.0	59.0
Model Section 2.2	*always In*, *always Out* *monotone increasing*	68.1	71.1	75.0	59.0

Notes: Inequity aversion, sequential reciprocity and guilt aversion modeled respectively as in Fehr and Schmidt (1999), Dufwenberg and Kirchsteiger (2004) and Battigalli and Dufwenberg (2007). Predictions for these models assume equilibrium play. For random play, inferences are based on $|P| = 11$ independent draws of a strategy from $\{In, Out\}$ for a sample of 113 A players, where $P = \{0, 0.1, ..., 1\}$ and $\mathbb{P}\{In\} = \mathbb{P}\{Out\} = \frac{1}{2}$. Outcome of the draw yielded: 1 *always In* (NI), 2 *monotone increasing* (1 in NI, 1 in COM), 4 *monotone decreasing* (1 in NI, 3 in COM) and 106 *non monotone*.

Table 2: % of B's behavior explained by other models (see Table 6 in main text)

Model	Rationalizable patterns	*NI*	*EI-In*	*EI-Out*	*COM-In*	*COM-Out*	*No-COM*
Random play	essentially *non monotone*	10.5	16.7	19.4	7.7	12.8	12.8
Selfish	*always Take*	39.5	41.7	55.6	35.9	48.7	41.0
Inequity aversion	*always Meet*, *always Take*	57.9	58.3	69.4	51.3	61.5	64.1
Sequential reciprocity	*always Take*, *monotone increasing* except in *EI-Out* and *COM-Out*	63.2	55.6	55.6	69.2	48.7	56.4
Guilt aversion*	*always Meet*, *always Take* *monotone increasing* in *EI* and *COM*	57.9	72.2	–	84.6	–	79.5
Model Section 2.2	*always Meet*, *always Take* in *All* *monotone increasing* in *All* except in *EI-Out* and *COM-Out*	81.6	72.2	69.4	84.6	61.5	79.5

Note: Inequity aversion, sequential reciprocity and guilt aversion modeled respectively as in Fehr and Schmidt (1999), Dufwenberg and Kirchsteiger (2004) and Battigalli and Dufwenberg (2007). Predictions for these models assume equilibrium play. For random play, inferences are based on $|P| = 11$ independent draws of a strategy from $\{In, Out\}$ for a sample of 227 B players, where $P = \{0, 0.1, ..., 1\}$ and $\mathbb{P}\{Meet\} = \mathbb{P}\{Take\} = \frac{1}{2}$. Outcome of the draw yielded: 2 *monotone increasing* (1 in *NI*, 1 in *COM-Out*), 2 *monotone decreasing* (1 in *EI-Out*, 1 in *No-COM*) and 223 *non monotone*. *No restrictions in *EI-Out* and *COM-Out*.

B Additional results and econometric analysis

B.1 Relationship between switching probabilities and theoretical threshold $\bar{\sigma}(p, \alpha, \theta)$

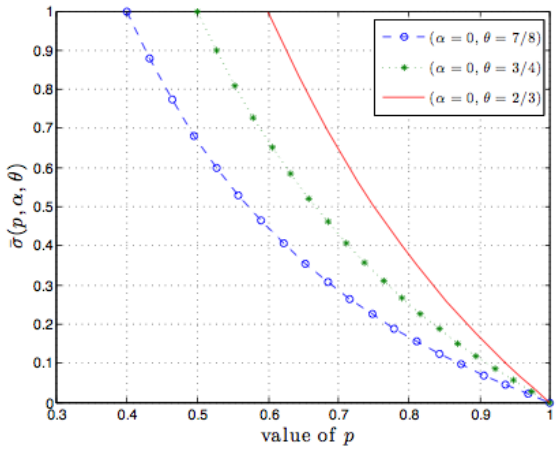
Table 3: Distribution of switching probabilities

value of the switching probability $p^* \geq \Lambda(\alpha, \theta)$	% of A players (N_A) *All* (1)	% of B players (N_B) *All* (2)	*Restricted* (3)
0.1	16.1 (5)	11.1 (5)	9.1 (3)
0.2	3.3 (1)	2.2 (1)	3.0 (1)
0.3	12.9 (4)	4.4 (2)	6.1 (2)
0.4	**16.1 (5)**	2.2 (1)	3.0 (1)
0.5	**35.5 (11)**	**24.4 (11)**	21.2 (7)
0.6	12.9 (4)	**35.6 (16)**	**36.4 (12)**
0.7	3.2 (1)	6.7 (3)	9.1 (3)
0.8	0.0 (0)	4.4 (2)	6.1 (2)
0.9	0.0 (0)	6.7 (3)	3.0 (1)
1	0.0 (0)	2.2 (1)	3.0 (1)
Total	100.0 (31)	100.0 (45)	100.0 (33)
Mean switching value	**0.403**	**0.538**	**0.539**
t-test	p-value	(1) vs (2): 0.005	(1) vs (3): 0.007

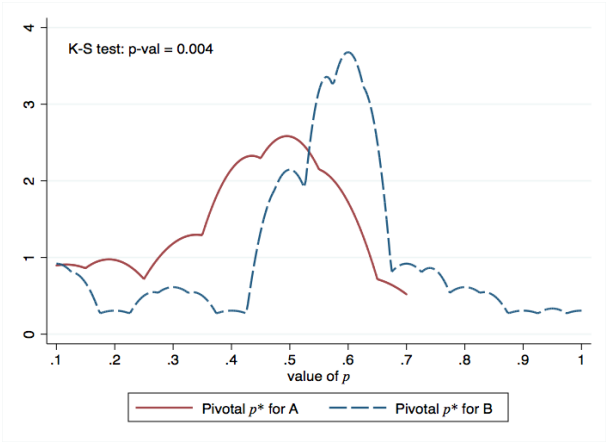
Notes: p^* corresponds to the value of p at which A (B) players switch from *Out* to *In* (*Take* to *Meet*), among those subjects whose cooperative response is strictly monotone increasing in p. Patterns of choice in *EI-In* and *EI-Out* are treated as coming from two separate B players. The three cases in *COM* are treated in a similar manner. *Restricted* excludes cases *EI-Out* and *COM-Out* for which the theory predicts no cooperation of B for any p. The two A players whose behavior is monotone increasing in p starting from $p = 0.1$ are included in the sample; results remain unchanged when excluding these subjects.

In the model presented in Section 2.2 of the main text, B's propensity to *Meet* (and therefore A's propensity to go *In*) depends on whether their belief σ_A^* (σ_A^{**}) is above the threshold $\bar{\sigma}(p, \alpha, \theta)$. Figure 1 Panel (a) displays the relationship between $\bar{\sigma}(p, \alpha, \theta)$ and p for different combinations of the parameters (α, θ). The combinations of (α, θ) were chosen by setting $\alpha = 0$ and finding the value of θ which generates the most frequent threshold probabilities $\Lambda(\alpha, \theta)$ observed in the data, 0.4, 0.5 and 0.6 (see Table 3 and Figure 1 Panel (b)).[19] In all three examples, this relationship is decreasing and convex, with stronger curvature as θ increases. As a result, in environments where A's decision is unknown, the relationship between prosocial behavior and p is increasing and concave in p. One implication is that the difference in B's prosociality between *NI* and *EI-In* (resp. *No-COM* and *COM-In*) should be highest right after the threshold $\Lambda(\alpha, \theta)$ and should vanish as p tends to 1.

[19] The probability p^* at which individuals switch to prosociality can only be identified with $\Lambda(\alpha, \theta)$ in *EI-In* and *COM-In*. In the other cases, $\Lambda(\alpha, \theta)$ serves as a lower bound for p^*.



(a) Relationship between $\bar{\sigma}(p, \alpha, \theta)$ and p



(b) Distribution of switching probabilities $p^* \geq \Lambda(\alpha, \theta)$

Figure 1: Relationship between prosociality, θ and p

B.2 Link between prosocial behavior and the value of p (see Section 4.1)

This section reports the results from the econometric analysis mentioned in Section 4.1 of the main text and presents some additional results. All baseline regressions are *OLS* regressions where the unit of observation is either a binary choice or a guess in {0,0.1,...,1} made by a given subject for a specific value of p and treatment case. Standard errors are clustered at the subject level to account for within-subject correlation across decisions. Because choices were inconsequential for A when $p = 0$ and for B when $p = 1$ in cases *EI-Out* and *COM-Out*, I report regression results both including and excluding these observations. Results are qualitatively similar. Only 9/113 subjects in role A exhibited a different behavior at $p = 0$ and none of the B players exhibited a different behavior at $p = 1$ in cases *EI-Out* and *COM-Out*. Thus, subjects seemed to have treated inconsequential decisions in a similar manner as decisions with a very low probability of occurrence; in other words, their strategies appear continuous in p.

B.2.1 Linear trends

Table 4 (resp. 5) examines whether a linear relationship exists between A's (resp. B's) propensity to choose *In* (resp. *Meet*) and the value of p. Although the coefficient has the right sign in all treatments, the linear trend is not significant for A, neither in the full nor in the restricted sample where observations for $p = 0$ are excluded. For B, the linear trend is positive and significant when pooling observations from all treatments. The breakdown by treatment case shows that this finding is essentially driven by behavior in *COM-In* and, to a lesser extent, by behavior in *COM-Out* (see Section 5.1 of the main text for a discussion of this latter finding, which is at odds with the theory).

Table 4: Effect of p on A's In Rate

Treatment	*All*		*NI*		*EI*		*COM*	
Sample	Full	$p \neq 0$	Full	$p \neq 0$	Full	$p \neq 0$	Full	$p \neq 0$
p	0.101	0.076	0.012	0.003	0.179	0.157	0.114	0.071
	[0.080]	[0.083]	[0.138]	[0.142]	[0.147]	[0.156]	[0.131]	[0.140]
Predicted effect of p	+		+		+		+	
Observations	1243	1130	418	380	396	360	429	390

Notes: Linear probability model with the dependent variable equal to 1 if A chose In regressed on the value of p $\in$ {0.1,...,1}. Full ($p \neq 0$) sample includes (excludes) observations for $p = 0$. Standard errors in square brackets clustered at the subject level.

Table 5: Effect of p on B's $Meet$ Rate

Treatment Case	*All*	*NI*	*No-COM*	*EI-In*	*COM-In*	*EI-Out*	*COM-Out*
Corresponding belief		$\sigma_A^* \in [0,1]$		$\sigma_A^* = 1$		$\sigma_A^* = 0$	
p	0.127**	0.177	-0.005	0.023	0.280**	-0.010	0.216*
	[0.052]	[0.115]	[0.104]	[0.113]	[0.115]	[0.080]	[0.112]
Predicted effect of p	+	+	+	+	+	0	0
Observations	2422	418	429	396	429	360	390

Notes: Linear probability model with the dependent variable equal to 1 if B chose $Meet$ regressed on the value of $p \in$ {0, 0.1,...,1}. Observations for $p = 1$ in *EI-Out* and *COM-Out* excluded from the analysis; results are almost identical on the full sample (coefficients are 0.023 [0.077] for *EI-Out*, 0.186* [0.102] *COM-Out* and 0.116** [0.049] for *All*). Standard errors in square brackets clustered at the subject level. * and ** indicate p-value < 0.1 and < 0.05.

B.2.2 Ordinal treatment of p

Without assuming a linear relationship, prosociality may be sensitive to whether p takes a low, medium or high value. Table 6 (resp. 7) shows the percentage of In (resp. $Meet$) choices for the following split of the data: $p_L \in \{0, 0.1, 0.2\}$, $p_M \in \{0.3, 0.4, 0.5, 0.6\}$ and $p_H \in \{0.7, 0.8, 0.9, 1\}$. Table 8 (resp. 9) presents a similar breakdown for alternative splits of the data at the cutoff values 0.3 and 0.7. Findings are only partially robust to alternative splits. For A, qualitative results remain unchanged but observed differences lose significance, except for treatment CI when comparing behavior at p_L and p_M. Results are robust for B in all treatment cases and, if anything, monotonicity is stronger with alternative splits.

Table 6: Percentage (freq.) of In choices by noise category

Treatment	*All*	*NI*	*EI*	*COM*
Noise category				
$p_L \in \{0, 0.1, 0.2\}$	**47.79**	48.25	**47.22**	**47.86**
	(162/339)	(55/114)	(51/108)	(56/117)
$p_M \in \{0.3, 0.4, 0.5, 0.6\}$	**55.97**	49.34	**59.03**	**59.62**
	(253/452)	(75/152)	(85/144)	(93/156)
$p_H \in \{0.7, 0.8, 0.9, 1\}$	**56.42**	50.00	**61.81**	57.69
	(255/452)	(76/152)	(89/144)	(90/156)
Predictions		t-Stat values		
$\sigma_{A,p_L} \leq \sigma_{A,p_M}$	2.04**	0.17	1.82**	1.64*
$\sigma_{A,p_L} \leq \sigma_{A,p_H}$	1.41*	0.14	1.31*	0.96
$\sigma_{A,p_M} \leq \sigma_{A,p_H}$	0.14	0.10	0.44	0.33

Notes: Linear probability model with the dependent variable equal to 1 if A chose In regressed on dummies for low, medium and high values of p, where $p_L \in \{0, 0.1, 0.2\}$, $p_M \in \{0.3, 0.4, 0.5, 0.6\}$ and $p_H \in \{0.7, 0.8, 0.9, 1\}$. Standard errors clustered at the subject level. Results qualitatively similar when excluding subjects with a different behavior at $p = 0$. Significance assessed with one-sided t-tests; * and ** indicate p-value < 0.1 and < 0.05.

Table 7: Percentage (freq.) of $Meet$ choices by noise category

Treatment Case	*All*	*NI*	*No-COM*	*EI-In*	*COM-In*	*EI-Out*	*COM-Out*
Noise category							
$p_L \in \{0, 0.1, 0.2\}$	**31.86**	**33.33**	42.74	34.36	**29.91**	24.07	**26.50**
	(217/681)	(38/114)	(50/117)	(37/108)	(35/117)	(26/108)	(31/117)
$p_M \in \{0.3, 0.4, 0.5, 0.6\}$	**32.93**	**36.84**	37.18	31.25	**35.26**	24.31	**32.05**
	(299/908)	(56/152)	(58/156)	(45/144)	(55/156)	(35/144)	(50/156)
$p_H \in \{0.7, 0.8, 0.9, 1\}$	**39.98**	**46.05**	41.03	37.50	**48.72**	25.00	**40.38**
	(363/908)	(70/152)	(64/156)	(54/144)	(76/156)	(36/144)	(63/156)
Predictions#			t-Stat values				
$\sigma_{B,p_L} \leq \sigma_{B,p_M}$	0.50	0.36	1.32*	0.69	1.02	0.10	1.35*
$\sigma_{B,p_L} \leq \sigma_{B,p_H}$	2.16**	1.43*	0.22	0.37	2.11**	0.17	1.81**
$\sigma_{B,p_M} \leq \sigma_{B,p_H}$	3.00***	1.80**	0.79	1.05	2.35**	0.17	1.84**

#Prediction is $\sigma_{B,p_L} = \sigma_{B,p_M} = \sigma_{B,p_H} = 0$ for *EI-Out* and *COM-Out*.

Notes: Linear probability model with the dependent variable equal to 1 if B chose $Meet$ regressed on dummies for low, medium and high values of p, where $p_L \in \{0, 0.1, 0.2\}$, $p_M \in \{0.3, 0.4, 0.5, 0.6\}$ and $p_H \in \{0.7, 0.8, 0.9, 1\}$. Standard errors clustered at the subject level. Results qualitatively similar when excluding observations for $p = 1$ in *EI-Out* and *COM-Out*. Significance assessed with one-sided t-tests; *, ** and *** indicate p-value < 0.1 , < 0.05 and <0.01.

Table 8: Percentage (freq.) of In choices by noise category for two alternative splits

Treatment	*All*	*NI*	*EI*	*COM*
Alternative split 1				
$p_L \in \{0, 0.1, 0.2, 0.3\}$	50.22	49.34	**48.61**	52.56
$p_M \in \{0.4, 0.5, 0.6\}$	55.46	48.25	**61.11**	57.26
$p_H \in \{0.7, 0.8, 0.9, 1\}$	56.42	50.00	61.81	57.69
Predictions				
$\sigma_{A,p_L} \leq \sigma_{A,p_M}$	1.16	0.14	1.56*	0.62
$\sigma_{A,p_L} \leq \sigma_{A,p_H}$	1.05	0.00	1.20	0.53
$\sigma_{A,p_M} \leq \sigma_{A,p_M}$	0.33	0.36	0.14	0.10
Alternative split 2				
$p_L \in \{0, 0.1, 0.2, 0.3\}$	50.22	49.34	**48.61**	52.56
$p_M \in \{0.4, 0.5, 0.6, 0.7\}$	55.75	48.68	**62.50**	56.41
$p_H \in \{0.8, 0.9, 1\}$	56.34	50.00	60.19	58.97
Predictions				
$\sigma_{A,p_L} \leq \sigma_{A,p_M}$	1.16	0.10	1.62*	0.48
$\sigma_{A,p_L} \leq \sigma_{A,p_H}$	1.02	0.00	1.04	0.66
$\sigma_{A,p_M} \leq \sigma_{A,p_M}$	0.22	0.32	0.47	0.59

Notes: Regression specification as in Table 6. Significance assessed with one-sided t-tests; * indicates p-value < 0.1.

Table 9: Percentage (freq.) of $Meet$ choices by noise category for two alternative splits

Treatment	*All*	*NI*	*No-COM*	*EI-In*	*COM-In*	*EI-Out*	*COM-Out*
Alternative split 1							
$p_L \in \{0, 0.1, 0.2, 0.3\}$	**31.39**	**32.89**	41.03	34.03	**28.85**	24.31	26.92
$p_M \in \{0.4, 0.5, 0.6\}$	**33.92**	**38.60**	37.61	30.56	**38.46**	24.07	33.33
$p_H \in \{0.7, 0.8, 0.9, 1\}$	**39.98**	**46.05**	41.03	37.50	**48.72**	25.00	40.38
Predictions#							
$\sigma_{B,p_L} \leq \sigma_{B,p_M}$	1.23	0.95	0.86	0.71	1.86**	0.00	1.45*
$\sigma_{B,p_L} \leq \sigma_{B,p_H}$	2.42***	1.56*	0.00	0.41	2.38**	0.14	1.82**
$\sigma_{B,p_M} \leq \sigma_{B,p_M}$	2.85***	1.68*	0.81	1.30	1.87**	0.24	1.85**
Alternative split 2							
$p_L \in \{0, 0.1, 0.2, 0.3\}$	**31.39**	**32.89**	41.03	34.03	**28.85**	24.31	26.92
$p_M \in \{0.4, 0.5, 0.6, 0.7\}$	**35.35**	**39.47**	38.46	34.72	**40.38**	22.22	35.90
$p_H \in \{0.8, 0.9, 1\}$	**40.09**	**47.37**	41.03	34.26	**49.57**	27.78	39.32
Predictions#							
$\sigma_{B,p_L} \leq \sigma_{B,p_M}$	1.69**	1.04	0.54	0.14	2.04**	0.55	1.79**
$\sigma_{B,p_L} \leq \sigma_{B,p_H}$	2.41***	1.64*	0.00	0.00	2.47***	0.60	1.65*
$\sigma_{B,p_M} \leq \sigma_{B,p_M}$	2.66***	1.76**	0.79	0.10	2.14**	1.45*	1.12

Notes: Regression specification as in Table 7. Significance assessed with one-sided t-tests; *, ** and *** indicate p-value < 0.1 , < 0.05 and <0.01. #Prediction is $\sigma_{B,p_L} = \sigma_{B,p_M} = \sigma_{B,p_H} = 0$ for *EI-Out* and *COM-Out*.

B.2.3 Comparison with other findings in the literature

Here I compare the results of this paper to previous findings in the literature. Despite being embedded in a more complex setting, observed choices in this paper are overall highly consistent with previous studies:

Table 10: Comparison with other papers

Paper	Expected payoffs (normalized)	Elicitation method	Treatment condition	A's % *In* of A players	% *Meet* of B players
Mc Cabe et al. (2003)	(5,5) if *Out*	direct response	$p = 0$ (NO COM)	—	33.3%
	(6.25, 6.25) if (*In*,*Meet*)	method	$p = 1$ (NO COM)	63.0%	64.7%
	(3.75, 7.5) if (*In*,*Take*)				
Cox and Deck (2006)	(5,5) if *Out*	direct response	$p = 0$ (NO COM)	—	35.0%
	(7.5, 12.5) if (*In*,*Meet*)	method	$p = \frac{3}{4}$ (NO COM)	33.9%	55.0%
	(0, 20) if (*In*,*Take*)		$p = 1$ (NO COM)	50.0%	63.6%
Charness and Dufwenberg (2006)	(5,5) if *Out*	strategy method	$p = 1$ (NO COM)	55.6%	44.4%
	(10, 10) if (*In*,*Meet*)		$p = 1$ (COM A→B)	67.4%	39.1%
	(0, 14) if (*In*,*Take*)				
This paper	(5,5) if *Out*	strategy method	$p = 0$ (*NI*)	47.4%	34.2%
	(10-$1_{\{C\}}$, 10) if (*In*,*Meet*)		$p \in \{0.7, 0.8\}$ (*NI*)	51.3%	43.4%
	(2-$1_{\{C\}}$, 14) if (*In*,*Take*)		$p = 1$ (*NI*)	47.4%	50.0%
			$p = 1$ (*COM*)	61.5%	51.3%*
				* % in *COM-In* for B	

Notes: Payoffs normalized so that *Out* payoffs are (5,5) in all games; normalizing factor = 4 in Mc Cabe et al. (2003), = 2 in Cox and Deck (2006) and = 1 in Charness and Dufwenberg (2006). In the latter, A's payoff from (*In*,*Meet*) is 12 with probability $\frac{5}{6}$ and 0 with probability $\frac{1}{6}$, yielding an expected payoff of 10. The indicator $1_{\{C\}}$ is equal to 1 if A chooses costly communication and *In* is realized. NO COM refers to the absence of communication between A and B, while COM A→B denotes one-sided (pre-play, cheap talk) communication from A to B.

For the A players, results when $p = 1$ are particularly close to those of Charness and Dufwenberg (2006), whose design is closest in terms of payoff structure and elicitation of strategies. The percentage of A players who choose *In* when $p = 1$ ranges from 47% in *NI* to 61% in both *EI* and *COM*; the corresponding numbers in Charness and Dufwenberg (2006) are 56% in their no communication treatment and 67% with one-sided communication from A to B. With somewhat different payoffs and the use of the direct response method in both cases, Cox and Deck (2006) and McCabe et al. (2003) respectively find that 50% and 63% of A players choose to trust in their binary trust game with no communication.

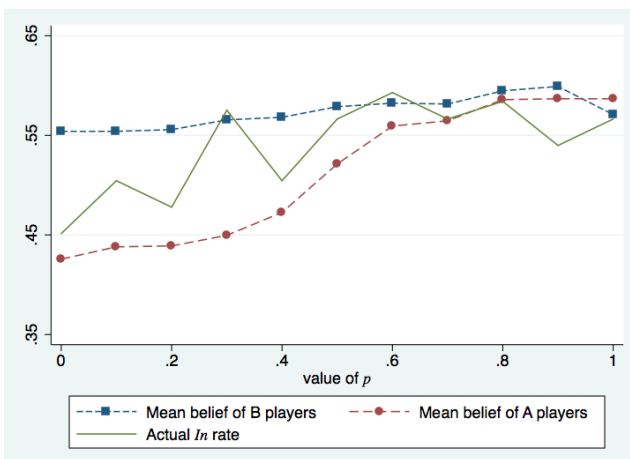
The behavior of the B players at the extreme values of p is also fairly consistent with previous findings. In the treatment which is closest to the studies of McCabe et al. (2003) or Cox and Deck (2006), *NI*, this experiment qualitatively reproduces their main finding: the percentage of B players who choose *Meet* is significantly higher for $p = 1$ than for $p = 0$ (50% versus 34%, p-value = 0.057

on a one-sided t-test).[20] Furthermore, behavior of the B players when $p = 0$ is almost identical across all three studies, with about one third of B players choosing the prosocial action. When $p = 1$, B's cooperation rate of 50% appears to be slightly smaller than in McCabe et al. (2003) and Cox and Deck (2006). This difference could be due to the use of the direct response method, if observing an act of trust triggers stronger emotional responses. Using the strategy method and the same payoffs for B as in the present study, Charness and Dufwenberg (2006) indeed find that 44% of B players choose the prosocial action in their baseline, no communication, treatment.

B.2.4 Relationship between beliefs and the value of p

In the main text, I show that the distribution of beliefs about A's propensity to go In changes as the value of p increases, with the mode of the distribution progressively shifting to 1 both for A and B. Furthermore, while B has more optimistic beliefs than A for low values of p, guesses become progressively more aligned as the value of p increases. Here I compare the first moment of each distribution, using the actual In rate as a benchmark. Pooling observations across all treatments, Figure 2 shows the mean first-order belief of B, $\bar{\sigma}_A^*$, the mean second-order belief of A, $\bar{\sigma}_A^{**}$, and the actual In rate for each value of p:

Figure 2: Average beliefs of A and B by value of p



Notes: For each value of p, the mean belief of the B players, $\bar{\sigma}_A^*$, corresponds to their average answer to the question: "How likely is it that A chose In?". In turn, the mean belief of the A players, $\bar{\sigma}_A^{**}$, corresponds to their average answer to the question: "What do you think was B's answer?". Observations pooled across all treatments.

[20] In addition, B's *Meet* rate when $p \in \{0.7, 0.8\}$ falls nominally between the rates at $p = 0$ and $p = 1$, finding which is in line with Cox and Deck (2006). On the other hand, A's In rate does not fall when $p \in \{0.7, 0.8\}$ relative to $p = 1$; if anything, it is higher. This latter finding contrasts with the results of Cox and Deck (2006).

The pattern of beliefs for B is almost flat: on average, the B players believe that A is 55.4% (resp. 57.1%) likely to choose In when $p = 0$ (resp. $p = 1$), a difference which is not statistically significant. Although the B players tend to slightly overestimate the likelihood that A chose In for low values of p, beliefs are fairly accurate: except for $p = 0$, the average belief $\bar{\sigma}_A^*$ is never statistically different from the actual In rate and the average percentage point deviation across values of p is only 3.8. While B's beliefs tend to be flat, A's beliefs exhibit a clear monotone increasing pattern: on average, A expects B to have answered that A was 42.6% (resp. 58.7%) likely to choose In for $p = 0$ (resp. $p = 1$), a difference which is statistically significant (p-value = 0.010 on a two-sided t-test).[21] While the A players tend to almost systematically underestimate B's beliefs, differences are only significant for $p < 0.5$ and are almost identical for $p > 0.5$. On the other hand, A's second-order beliefs are very close to the actual In rate. The percentage point deviation of $\bar{\sigma}_A^{**}$ from the actual In rate averages 4.0 across all values of p.[22]

Table 11 shows that the positive trend followed by A's beliefs σ_A^{**} is statistically significant in the full sample and is mostly due to a strong monotonicity in A's beliefs in COM. On the other hand, there is no such a relationship for B in any treatment.

Table 11: Effect of p on players' beliefs

Treatment	*All*	*NI*	*CI*	*COM*
	Belief σ_A^* of B players			
p	0.039	0.075	-0.016	0.055
	[0.053]	[0.089]	[0.078]	[0.105]
	Belief σ_A^{**} of A players			
p	0.204***	0.150	0.202	0.259***
	[0.065]	[0.120]	[0.127]	[0.093]
Predicted effect of p	+	+	+	+
Observations	1130	380	360	390

The dependent variable is B's elicited belief $\sigma_A^* \in \{0, 0.1, ..., 1\}$ that A chose In and A's elicited belief $\sigma_A^{**} \in \{0, 0.1, ..., 1\}$ about B's belief. Standard errors in square brackets are clustered at the subject level. Beliefs formed for $p = 0$ excluded from the analysis; results qualitatively similar when including all observations.

[21] A regression of σ_A^{**} on p in the full sample shows that the linear trend is strongly statistically significant ($p < 0.01$).

[22] The consistency between A's behavior and beliefs is somewhat less surprising as it could simply reflect a false consensus bias i.e. the tendency to think that others can predict one's own behavior well. See Vanberg (2008), Ellingsen et al. (2010) or Butler et al. (2013) for a discussion of the false consensus bias in experimental trust games.

B.2.5 Robustness analysis

Here I present results about the distribution of Trust and No Trust profiles (Table 2 in the main text) and percentage of *Meet* choices in *COM* (Table 4 in the main text) for an alternative split of the data i.e. including the cutoff of $p = 0.5$ in the high values of p. Results are qualitatively unchanged with this different split of the values of p into high and low categories.

Table 12: % (freq.) of the Trust and No Trust profiles by treatment and value of p

Action profile	(*In*, *Meet*)				(*Out*, *Take*)			
Treatment	*All*	*NI*	*EI*	*COM*	*All*	*NI*	*EI*	*COM*
$p < 0.5$	**15.49**	**13.16**	15.28	17.95	**31.42**	32.02	**34.72**	**30.13**
	(70/452)	(20/152)	(22/144)	(28/156)	(142/452)	(45/152)	(50/144)	(47/156)
	$\wedge^{***}$	$\wedge^{**}$	$\wedge^{*}$	$\wedge$	$\vee^{*}$	$\vee$	$\vee^{*}$	$\vee^{*}$
$p \geq 0.5$	**24.25**	**25.00**	25.25	22.65	**24.55**	**29.61**	**21.78**	**19.66**
	(161/664)	(57/228)	(51/202)	(53/234)	(163/664)	(73/228)	(44/202)	(46/234)

Notes: Results from a linear probability model with the dependent variable equal to 1 if an individual pair played the profile (*In*, *Meet*) (respectively, (*Out*, *Take*)) and 0 otherwise, regressed on dummies for $p < 0.5$ and $p \geq 0.5$. Realized action profiles for $p = 0$ dropped from all treatments; action profiles for $p = 1$ dropped in *EI* and *COM* when A chose *Out*. Standard errors clustered at the level of each pair. Significance assessed with one-sided t-tests; * and ** indicate p-value <0.1 and < 0.05.

Table 13: Percentage of *Meet* choices in *COM* by case and value of p

	$p < 0.5$			$p \geq 0.5$		
Column	(1)	(2)	(3)	(4)	(5)	(6)
Treatment Case	*COM-In*	*COM-Out*	*No-COM*	*COM-In*	*COM-Out*	*No-COM*
% of *Meet*	29.23	26.15	40.00	46.58	40.00	40.17
t-Stat	0.52	2.00**	1.31*	0.98	0.00	1.35*
Comparison	(1) vs (2)	(2) vs (3)	(1) vs (3)	(4) vs (5)	(5) vs (6)	(4) vs (6)
Predicted sign	(1) $\geq$ (2)	(2) $\leq$ (3)	(1) $\geq$ (3)	(4) $\geq$ (5)	(5) $\leq$ (6)	(4) $\geq$ (6)
Observations	195	195	195	234	195	234

Notes: Results from a linear probability model with the dependent variable equal to 1 if B chose *Meet* and 0 otherwise, regressed on dummies for the treatment case, dividing the sample into $p < 0.5$ and $p \geq 0.5$. Observations for $p = 1$ excluded in *COM-Out*; results essentially unchanged when including all observations. Standard errors clustered at the subject level. Significance is assessed with one-sided t-tests. * and ** indicate p-value < 0.1 and < 0.05.

Following the spirit of Table 2 & 12, I also show in Table 14 the distribution of equilibrium profiles (including communication choices by A) for low and high values of p; the equilibrium of Trust with signalling (No Trust with no signalling) is significantly more (less) likely to emerge for high values of p.

Table 14: % (freq.) of equilibrium profiles in COM by value of p

	action profile		
value of p	$(In, C; Meet)$	$(In, NC; Meet)$	$(Out, NC; Take)$
$p \leq 0.5$	**3.6**	14.4	**27.7**
	(7/195)	(28/195)	(54/195)
	$\wedge^{**}$		$\vee^{**}$
$p > 0.5$	**12.3**	11.3	**16.9**
	(24/195)	(22/195)	(33/195)

Results from a linear probability model with dependent variable equal to 1 if an individual pair played the profile (In, C; $Meet$) (respectively, (In, NC; $Meet$) or (Out, NC; $Take$)) and 0 otherwise, regressed on dummies for $p \leq 0.5$ and $p > 0.5$; results qualitatively unchanged when splitting the data at $p < 0.5$ and $p \geq 0.5$. Standard errors clustered at the level of each pair. Significance assessed with one-sided t-tests; ** indicates p-value < 0.05.

B.3 A's optimal strategy given B's actual play (see 4.2-4.3 in the main text)

In this section, I compute A's optimal strategy given the fraction of B players who $Meet$ for each value of p and treatment.

Table 15: B's actual play by treatment case

	Proportion σ_B of B players who choose $Meet$ in:					
value of p	*NI*	*EI-In*	*EI-Out*	*COM-In*	*COM-Out*	*No-COM*
0.1	0.342	0.361	0.222	0.282	0.308	0.436
0.2	0.316	0.333	0.194	0.333	0.256	0.436
0.3	0.316	0.333	0.250	0.256	0.282	0.359
0.4	0.395	0.250	0.222	0.308	0.231	0.359
0.5	0.342	0.333	0.250	0.308	0.385	0.385
0.6	0.421	0.333	0.250	0.538	0.385	0.385
0.7	0.421	0.472	0.167	0.462	0.436	0.410
0.8	0.447	0.417	0.250	0.487	0.359	0.410
0.9	0.474	0.306	0.278	0.487	0.436	0.410
1	0.5	0.306	0.306	0.513	0.385	0.410

Treatment NI: Let $\sigma_B \in [0,1]$ refer to the proportion of B players who choose $Meet$. Choosing In is optimal for A if and only if

$$[p + \frac{1}{2}(1-p)][10\sigma_B + 2(1-\sigma_B)] + \frac{1}{2}(1-p) \cdot 5 \geq \frac{1}{2}(1-p)[10\sigma_B + 2(1-\sigma_B)] + [p + \frac{1}{2}(1-p)] \cdot 5$$

$$\Leftrightarrow \sigma_B \geq 0.375$$

This condition is satisfied in the data for $p = 0.4$ and $p \geq 0.6$. In the other cases, A's optimal action is therefore to choose Out.

Treatment EI: Here B can adopt a strategy which conditions on the action taken by A. Let $\sigma_{B|In} \in [0,1]$ refer to the proportion of B players who choose $Meet$ given that A chose In and similarly define $\sigma_{B|Out} \in [0,1]$. Choosing In is optimal for A if and only if

$$[p+\frac{1}{2}(1-p)][10\sigma_{B|In}+2(1-\sigma_{B|In})]+\frac{1}{2}(1-p)\cdot 5 \geq \frac{1}{2}(1-p)[10\sigma_{B|Out}+2(1-\sigma_{B|Out})]+[p+\frac{1}{2}(1-p)]\cdot 5$$

$$\Leftrightarrow \sigma_{B|In} \geq \frac{3p + 4(1-p)\sigma_{B|Out}}{4(1+p)}$$

Given B's actual play in EI, it can be checked that A's optimal choice is to choose In for $p \in \{0.1, 0.2, 0.3, 0.5, 0.7, 0.8\}$ and Out otherwise. These findings are summarized in the table below and compared against A's actual play.

Table 16: Optimal versus actual behavior of A in NI and EI

	optimal strategy of A given B's actual play in:		% optimal play of A given B's actual play in:	
value of p	NI	EI	NI	EI
0.1	*Out*	*In*	52.6	50.0
0.2	*Out*	*In*	52.6	47.2
0.3	*Out*	*In*	50.0	52.8
0.4	*In*	*Out*	52.6	47.2
0.5	*Out*	*In*	50.0	63.9
0.6	***In***	***Out***	**47.4**	**33.3**
0.7	*In*	*In*	50.0	66.7
0.8	*In*	*In*	52.6	63.9
0.9	*In*	***Out***	50.0	**44.4**
1	***In***	***Out***	**47.4**	**38.9**

Treatment COM: In this last treatment, B can adopt a different strategy depending on whether A chose to communicate In, A chose to communicate Out or A chose No Communication. Let $\mathcal{H} := \{In, Out, NC\}$ be the set of corresponding histories, where $NC := In \sqcup Out$, and denote by $\sigma_{B|h}$ the proportion of B players who $Meet$ after history h. It is first easy to show that playing (Out, C) is never optimal for A given B's actual play. In particular, (Out, NC) always gives higher expected payoffs to A, as the proportion of B players who $Meet$ is either almost identical in COM-Out and No-COM or strictly higher in No-COM, and signalling is costly.[23] Therefore, depending on B's choice for a specific value of p, 3 strategies can be optimal for A i.e. (In, C), (In, NC) and (Out, NC).

1- A will prefer (In, C) over (In, NC) if and only if

$$[p+\frac{1}{2}(1-p)][9\sigma_{B|In}+(1-\sigma_{B|In})]+\frac{1}{2}(1-p)\cdot 5 \geq [p+\frac{1}{2}(1-p)][10\sigma_{B|NC}+2(1-\sigma_{B|NC})]+\frac{1}{2}(1-p)\cdot 5$$

$$\Leftrightarrow \ \sigma_{B|In} - \sigma_{B|NC} \geq 0.125 \tag{6}$$

2- A will prefer (In, C) over (Out, NC) if and only if

$$[p+\frac{1}{2}(1-p)][9\sigma_{B|In}+(1-\sigma_{B|In})]+\frac{1}{2}(1-p)\cdot 5 \geq \frac{1}{2}(1-p)[10\sigma_{B|NC}+2(1-\sigma_{B|NC})]+[p+\frac{1}{2}(1-p)]\cdot 5$$

$$\Leftrightarrow \ \sigma_{B|In} \geq \frac{7p+1+8(1-p)\sigma_{B|NC}}{8(1+p)} \tag{7}$$

3- Finally, A will prefer (In, NC) over (Out, NC) if and only if

$$[p+\frac{1}{2}(1-p)][10\sigma_{B|NC}+2(1-\sigma_{B|NC})]+\frac{1}{2}(1-p)\cdot 5 \geq \frac{1}{2}(1-p)[10\sigma_{B|NC}+2(1-\sigma_{B|NC})]+[p+\frac{1}{2}(1-p)]\cdot 5$$

$$\Leftrightarrow \ \sigma_{B|NC} \geq 0.375 \tag{8}$$

Combining conditions (1)-(3), it is easy to verify that (In, C) is optimal for $p = 0.6$, (Out, NC) for $p \in \{0.3, 0.4\}$ and (In, NC) for all the other values. The following table compares A's optimal and actual play in COM.

[23] For A to prefer (Out, C) over (Out, NC), it must be the case that

$$\frac{1}{2}(1-p)[9\sigma_{B|Out}+(1-\sigma_{B|Out})]+[p+\frac{1}{2}(1-p)]\cdot 5 \geq \frac{1}{2}(1-p)[10\sigma_{B|NC}+2(1-\sigma_{B|NC})]+[p+\frac{1}{2}(1-p)]\cdot 5$$

$$\Leftrightarrow \ \sigma_{B|Out} \geq \sigma_{B|NC} + 0.125$$

This condition is never satisfied in the data for any value of p.

Table 17: Optimal versus actual behavior of A in COM

value of p	optimal strategy of A given B's actual play	% optimal play of A given B's actual play
0.1	(In, NC)	35.9
0.2	(In, NC)	33.3
0.3	(Out, NC)	33.3
0.4	(Out, NC)	46.2
0.5	(In, NC)	35.9
0.6	$\boldsymbol{(In, C)}$	**25.6**
0.7	(In, NC)	25.6
0.8	(In, NC)	30.8
0.9	(In, NC)	23.1
1	(In, NC)	28.2

C Instructions

C.1 Instructions for the baseline treatment (NI)

You are about to participate in an experiment on decision-making. You will receive 5 dollars for your participation, irrespective of your decisions. You may also receive additional earnings depending partly on your decisions, partly on the decisions of others, and partly on chance. You will be paid with a cash voucher, privately at the end of the session.

Please turn off cellular phones and similar devices now. All interactions between you and other participants will take place through your computer terminals. Please do not talk or in any way try to communicate with other participants during the session. If you have anything on your table, please place it back in your bag.

During the instruction period, you will be given a description of the main features of the session and will be shown the computer interface. If you have any questions during the period, raise your hand and your question will be answered so everyone can hear.

The experiment has 2 parts; we will start with instructions for part 1. Once this part is over, instructions for part 2 will be distributed to you. Your decisions in part 1 will have no impact on your earnings in part 2.

Instructions for Part 1

All participants will be randomly assigned a role, either A or B, for the entire session (part 1 and 2). You will interact with one participant in the other role; the participant you are matched with will be selected at random.

Participants will in turn make a choice between two options:

- First, participants in role A will choose between In or Out.

- Second, participants in role B will choose between Up or $Down$.

If A chooses Out, the interaction ends. Both A and B receive 5 dollars.

If A chooses In, payoffs depend on B's decision:

- If B chooses Up, A receives 2 dollars and B receives 14 dollars.

- If B chooses $Down$, both A and B receive 10 dollars.

However, before participants make a decision, **a computer will randomly determine whether A's decision is implemented or not**:

- If A's decision is implemented, then this is the final decision faced by B.

- If A's decision is not implemented, then the final decision faced by B will be determined by a random choice of the computer.

The computer will select In and Out with equal chance. More precisely, the computer will simulate a coin toss: if heads comes up, In will be selected while if tails comes up, Out will be selected.

B will be asked to make a choice without knowing whether In was the final decision. However, since B's decision can only impact payoffs if In was indeed the final decision, B should make a choice accordingly.

The percentage chance p that A's decision gets implemented will be randomly selected among the options **0, 10, 20, 30, 40, 50, 60, 70, 80, 90 or 100%.** For instance:

- If $p = 0$, then B will face the computer's choice for sure.
- If $p = 100$, then B will face A's decision for sure.
- If $p = 40$, then B will face A's decision with a 40% chance and the computer's choice with a 60% chance.

Note that for any value of p between 10 and 90, assuming In was the final decision, B will not know whether A chose In and A's decision was implemented OR the computer chose In and replaced A's decision.

However, note that **B is more likely to face A's decision as the value of p increases**: when p is small, the final decision faced by B is more likely to come from the computer while when p is large, it is more likely to come from A.

In order to determine whether A's decision gets implemented, the computer will randomly draw a number between 1 and 100. For instance, if p is 70%, then:
- A's decision will be implemented if the computer draws any number from 1 to 70 included.
- A's decision will not be implemented if the computer draws any number from 71 to 100 included.

You will not know the value of p while making your decision. Instead, we will ask you to **make a choice for each possible value of p between 0 and 100**. Thus, if you are in role A (resp. role B), you will make a choice between In or Out (resp. Up or $Down$) for each possible percentage chance that your (resp. A's) decision gets implemented.

After the decisions have been made, the computer will randomly select the value of p that will count for payment. All values have equal chance to be selected. For instance, suppose that the randomly selected value of p is 20, meaning A's decision is implemented with a 20% chance. Then A and B will be paid according to their choice for the case corresponding to $p = 20$.

You will not receive any feedback about actions or earnings between part 1 and 2. At the end of the experiment, you will only know how much you earned in total (part 1 and 2).

To summarize:

STEP 1: A makes a choice between *In* or *Out* for each possible value of p, which is the percentage chance that her decision gets implemented.

STEP 2: B makes a choice between *Up* and *Down* for each possible value of p.

STEP 3: Finally, payoffs are determined according to the selected value of p:

1. If A's decision is implemented, payoffs are determined according to A's and (possibly) B's choice for the selected value of p:

2. If A chooses *Out*, both A and B get 5 dollars, irrespective of B's decision.

   (a) If A chooses *In*, payoffs depend on B's decision: if B chooses *Up*, A gets 2 dollars and B gets 14 dollars; if B chooses *Down*, both A and B get 10 dollars.

3. If A's decision is not implemented, payoffs are determined according to the computer's choice and (possibly) B's choice for the selected value of p:

   (a) If the computer selects *Out*, both A and B get 5 dollars, irrespective of B's decision.

   (b) If the computer selects *In*, payoffs depend on B's decision: if B chooses *Up*, A gets 2 dollars and B gets 14 dollars; if B chooses *Down*, both A and B get 10 dollars.

Information about decisions and earnings is summarized in the table below:

	A gets	B gets
If *Out* is the final decision (BULLET 1(a) or 2(a))	5	5
If *In* is the final decision (BULLET 1(b) or 2(b)) and B chooses *Up*	2	14
If *In* is the final decision (BULLET 1(b) or 2(b)) and B chooses *Down*	10	10

Instructions for Part 2

We will now ask you to make guesses about the choices or beliefs of the participant you interacted with during part 1.

IF YOU ARE IN ROLE B:

We will ask you to tell us for each of the possible values of p, **how likely you think A chose *In*.** Thus, you will answer 11 questions in total, one for each value of p between 0 and 100.

You will be paid according to your answer to the **question which corresponds to the realized value of p**. For instance, if the selected p was 80, you will be paid according to your answer to the question: "How likely do you think A chose *In* if her decision is implemented with a 80%

chance?". You can earn 5 dollars for your answer to this question.

To determine your payment, we use a method under which you maximize your chances of winning 5 dollars by submitting what you truly believe are the chances that A chose In. More precisely, your compensation will be determined as follows:

1. You will first tell us how likely you think A chose In by picking a number among the options $\{0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$. For example, if you pick 60, this means that you think there is a 60% chance that A chose In.

2. Second, the computer will randomly draw a number between 0 and 100.

3. If the random number is equal or smaller than the number you submitted, then you will receive 5 dollars if A indeed chose In. For example, say that the random number is 24 and that you thought A was 60% likely to choose In. Because the number you submitted (60) is higher than the random number (24), you will receive 5 dollars if A indeed chose In and 0 if she chose Out.

4. If the random number is higher than the number you submitted, then you will receive 5 dollars with the probability determined by the random number. In other words, in that case, the chances that you receive 5 dollars are equal to the random number. For example, say that you thought A was 80% likely to choose In and the random number picked by the computer is 88; because the number you submitted (80) is lower than the random number (88), you will receive 5 dollars with a 88% chance.

   What should you do when answering the question: "How likely do you think A chose In"? The best you can do is to choose the number that is closest to how likely you think A chose In. Indeed, suppose for example that you believe there is a 60% chance that A chose In but you choose to select 40. Then you will regret your choice if the computer for instance selects 50. Since the random number (50) is higher than the number you submitted (40), you will earn 5 dollars with a 50% chance. On the other hand, if you had declared 60, you would have earned 5 dollars in case A indeed chose In, which you think has a 60% chance to occur.

If you are in role A:

The B you were matched with was asked to tell us how likely he thinks you chose In for each of the possible values of p, making 11 questions in total. For each question, B could choose a number among the options $\{0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$. For example, if B picked 30 for $p = 40$, this means that B thought you were 30% likely to choose In if your decision is implemented with a 40% chance.

For each of the 11 questions, we ask you to **guess B's answer**. You will be paid 5 dollars if you guess correctly B's answer to the **question which corresponds to the realized value of p**. For instance, suppose that the realized value of p is 40 and that B picked 30 for this case. Then you will receive 5 dollars if you choose 30 and nothing for any other number.

C.2 Part 1 of the instructions for treatment EI (see NI for Part 2)

You are about to participate in an experiment on decision-making. You will receive 5 dollars for your participation, irrespective of your decisions. You may also receive additional earnings depending partly on your decisions, partly on the decisions of others, and partly on chance. You will be paid with a cash voucher, privately at the end of the session.

Please turn off cellular phones and similar devices now. All interactions between you and other participants will take place through your computer terminals. Please do not talk or in any way try to communicate with other participants during the session. If you have anything on your table, please place it back in your bag.

During the instruction period, you will be given a description of the main features of the session and will be shown the computer interface. If you have any questions during the period, raise your hand and your question will be answered so everyone can hear.

The experiment has 2 parts; we will start with instructions for part 1. Once this part is over, instructions for part 2 will be distributed to you. Your decisions in part 1 will have no impact on your earnings in part 2.

Part 1: General Instructions

All participants will be randomly assigned a role, either A or B, for the entire session (part 1 and 2). You will interact with one participant in the other role; the participant you are matched with will be selected at random.

Participants will make a choice in sequence between two options:

- First, participants in role A will choose between *In* or *Out*.

- Second, participants in role B will choose between *Up* or *Down*.

If A chooses *Out*, the interaction ends. Both A and B receive 5 dollars.

If A chooses *In*, payoffs depend on B's decision:

- If B chooses *Up*, A receives 2 dollars and B receives 14 dollars.

- If B chooses *Down*, both A and B receive 10 dollars.

However, before participants make a decision, **a computer will randomly determine whether A's decision is implemented or not**.

- If A's decision is implemented, then this is the final decision faced by B.

- If A's decision is not implemented, then the final decision faced by B will be determined by a random choice of the computer.

  The computer will select In and Out with equal chance. More precisely, the computer will simulate a coin toss: if heads comes up, In will be selected while if tails comes up, Out will be selected.

B will be asked to make a choice for each of the following two cases:

- CASE 1: The final decision was In and A chose In.

  In this case, either A's decision was implemented or the computer drew In and replaced A's decision.

- CASE 2: The final decision was In and A chose Out.

  Note that in this case, the computer must have drawn In and replaced A's decision.

If CASE 1 occurs, then payments will be determined by B's choice in CASE 1.
If CASE 2 occurs, then payments will be determined by B's choice in CASE 2.

B will be asked to make a choice without knowing whether In was the final decision, or whether A chose In (CASE 1) or A chose Out (CASE 2). However, since B's decision in either case can only impact payoffs if this case occurs, B should choose accordingly.

The percentage chance p that A's decision gets implemented will be randomly selected among the options **0, 10, 20, 30, 40, 50, 60, 70, 80, 90 or 100%.** For instance:

- If $p = 0$, then B will face the computer's choice for sure.

- If $p = 100$, then B will face A's decision for sure.

- If $p = 40$, then B will face A's decision with a 40% chance and the computer's choice with a 60% chance.

Notice that **B is more likely to face A's decision as the value of p increases**: when p is small, the final decision faced by B is more likely to come from the computer while when p is large, it is more likely to come from A.

In order to determine whether A's decision gets implemented, the computer will randomly draw a number between 1 and 100. For instance, if p is 70%, then:

- A's decision will be implemented if the computer draws any number from 1 to 70 included.

- A's decision will not be implemented if the computer draws any number from 71 to 100 included.

You will not know the value of p while making your decision. Instead, we will ask you to **make a choice for each possible value of p between 0 and 100**. Thus, if you are in role A (resp. role B), you will make a choice between In or Out (resp. Up or $Down$) for each possible percentage chance that your (resp. A's) decision gets implemented.

After the decisions have been made, the computer will randomly select the value of p that will count for payment. All values have equal chance to be selected. For instance, suppose that the randomly selected value of p is 20, meaning A's decision is implemented with a 20% chance. Then A and B will be paid according to their choice for the case corresponding to $p = 20$.

You will not receive any feedback about actions or earnings between part 1 and 2. At the end of the experiment, you will only know how much you earned in total (part 1 and 2).

To summarize:

STEP 1: A makes a choice between In or Out for each possible value of p, which is the percentage chance that her decision gets implemented.

STEP 2: B makes a choice between Up and $Down$ for each possible value of p and for each of the following two cases:

- CASE 1: The final decision was In and A chose In.
- CASE 2: The final decision was In and A chose Out.

STEP 3: Finally, payoffs are determined according to the selected value of p:

1. If A's decision is implemented, payoffs are determined according to A's and (possibly) B's choice for the selected value of p:
   (a) If A chooses Out, both A and B get 5 dollars, irrespective of B's decision.
   (b) If A chooses In, payoffs depend on B's choice in CASE 1: if B chooses Up, A gets 2 dollars and B gets 14 dollars; if B chooses $Down$, both A and B get 10 dollars.
2. If A's decision is not implemented, payoffs are determined according to the computer's choice and (possibly) B's choice for the selected value of p:
   (a) If the computer selects Out, both A and B get 5 dollars, irrespective of B's decision.
   (b) If the computer selects In, payoffs depend on B's decision in CASE 1 (CASE 2) if A chose In (chose Out): if B chooses Up, A gets 2 dollars and B gets 14 dollars; if B chooses $Down$, both A and B get 10 dollars.

Information about decisions and earnings is summarized in the table below:

	A gets	B gets
If *Out* is the final decision (BULLET 1(a) or 2(a))	5	5
If *In* is the final decision (BULLET 1(b) or 2(b)) and B chooses *Up*	2	14
If *In* is the final decision (BULLET 1(b) or 2(b)) and B chooses *Down*	10	10

C.3 Instructions for treatment *COM* (Part 1 and 2)

You are about to participate in an experiment on decision-making. You will receive 7 dollars for your participation, irrespective of your decisions. You may also receive additional earnings depending partly on your decisions, partly on the decisions of others, and partly on chance. You will be paid with a cash voucher, privately at the end of the session.

Please turn off cellular phones and similar devices now. All interactions between you and other participants will take place through your computer terminals. Please do not talk or in any way try to communicate with other participants during the session. If you have anything on your table, please place it back in your bag.

During the instruction period, you will be given a description of the main features of the session and will be shown the computer interface. If you have any questions during the period, raise your hand and your question will be answered so everyone can hear.

The experiment has 2 parts; we will start with instructions for part 1. Once this part is over, instructions for part 2 will be distributed to you. Your decisions in part 1 will have no impact on your earnings in part 2.

Part 1: General Instructions

All participants will be randomly assigned a role, either A or B, for the entire session (part 1 and 2). You will interact with one participant in the other role; the participant you are matched with will be selected at random.

Participants will make a choice in turn between two options:

- First, participants in role A will choose between *In* or *Out*.
- Second, participants in role B will choose between *Up* or *Down*.

If A chooses *Out*, the interaction ends. Both A and B receive 5 dollars.

If A chooses *In*, payoffs depend on B's decision:

- If B chooses *Up*, A receives 2 dollars and B receives 14 dollars.
- If B chooses *Down*, both A and B receive 10 dollars.

However, before participants make a decision, **a computer will randomly determine whether A's decision is implemented or not**.

- If A's decision is implemented, then this is the final decision faced by B.
- If A's decision is not implemented, then the final decision faced by B will be determined by a random choice of the computer.

The computer will select *In* and *Out* with equal chance. More precisely, the computer will simulate a coin toss: if heads comes up, *In* will be selected while if tails comes up, *Out* will be selected.

Before B makes a decision, A will be offered the additional option to **inform B of her actual decision at a cost of 1 dollar**, which will be subtracted from A's final payoffs. Importantly, A will only have the option to reveal the choice that she really made. Therefore, there are 3 cases:

- CASE 1: *In* was the final decision, A chose *In* and paid to inform B that she chose *In*.
- CASE 2: *In* was the final decision, A chose *Out* and paid to inform B that she chose *Out*.
- CASE 3: *In* was the final decision and A didn't pay to inform B of her actual decision. In this case, she might have chosen *In* or *Out*.

Notice that in CASE 1 or CASE 3, either A's decision was implemented or the computer drew *In* and replaced A's choice. In CASE 2, the computer must have drawn *In* and replaced A's choice.

B will be asked to make a choice between *Up* and *Down* for each of the 3 cases:

- If CASE 1 occurs, then payments will be determined according to B's choice in CASE 1.
- If CASE 2 occurs, then payments will be determined according to B's choice in CASE 2.
- If CASE 3 occurs, then payments will be determined according to B's choice in CASE 3.

B will make a choice without knowing whether *In* was the final decision and without knowing the decisions made by A (corresponding to CASE 1, CASE 2 or CASE 3). However, since B's decision in a given case can only impact payoffs if this case occurred, B should choose accordingly.

The percentage chance p that A's decision gets implemented will be randomly selected among the options **0, 10, 20, 30, 40, 50, 60, 70, 80, 90 or 100%.** For instance:

- If $p = 0$, then B will face the computer's choice for sure.
- If $p = 100$, then B will face A's decision for sure.
- If $p = 40$, then B will face A's decision with a 40% chance and the computer's choice with a 60% chance.

Notice that **B is more likely to face A's decision as the value of p increases**: when p is small, the final decision faced by B is more likely to come from the computer while when p is large,

it is more likely to come from A.

In order to determine whether A's decision gets implemented, the computer will randomly draw a number between 1 and 100. For instance, if p is 70%, then:

- A's decision will be implemented if the computer draws any number from 1 to 70 included.
- A's decision will not be implemented if the computer draws any number from 71 to 100 included.

You will not know the value of p while making your decision. Instead, we will ask you to **make a choice for each possible value of p between 0 and 100**. Thus, if you are in role A (resp. role B), you will make a choice between *In* or *Out* (resp. *Up* or *Down*) for each possible percentage chance that your (resp. A's) decision gets implemented. After the decisions have been made, the computer will randomly select the value of p that will count for payment. All values have equal chance to be selected. For instance, suppose that the randomly selected value of p is 20, meaning A's decision is implemented with a 20% chance. Then A and B will be paid according to their choice for the case corresponding to $p = 20$.

You will not receive any feedback about actions or earnings between part 1 and 2. At the end of the experiment, you will only know how much you earned in total (part 1 and 2).

To summarize:

STEP 1: A makes a choice between *In* or *Out* for each possible value of p, which is the percentage chance that her decision gets implemented.

STEP 2: A chooses whether to pay to inform B of her choice for each possible value of p.

STEP 3: B makes a choice between *Up* and *Down* for each value of p and each of the following 3 cases:

- CASE 1: The final decision was *In*, A chose *In* and paid to inform B that she chose *In*.
- CASE 2: The final decision was *In*, A chose *Out* and paid to inform B that she chose *Out*.
- CASE 3: The final decision was *In* and A didn't pay to inform B of her choice.

STEP 4: Finally, payoffs are determined according to the selected value of p:

1. If A's decision is implemented, payoffs are determined according to A's and (possibly) B's choice for the selected value of p:

   (a) If A chooses *Out*, both A and B get 5 dollars, irrespective of B's decision.

(b) If A chooses *In*, payoffs depend on B's choice in CASE 1 or CASE 3 depending on A's choice: if B chooses *Up*, A gets 2 dollars (- 1 dollar in CASE 1) and B gets 14 dollars; if B chooses *Down*, both A and B get 10 dollars (- 1 dollar for A in CASE 1).

2. If A's decision is not implemented, payoffs are determined according to the computer's choice and (possibly) B's choice for the selected value of p:

   (a) If the computer selects *Out*, both A and B get 5 dollars, irrespective of B's decision.

   (b) If the computer selects *In*, payoffs depend on B's decision in CASE 1, CASE 2 or CASE 3 depending on A's choice: if B chooses *Up*, A gets 2 dollars (- 1 dollar in CASE 1 or 2) and B gets 14 dollars; if B chooses *Down*, both A and B get 10 dollars (- 1 dollar for A in CASE 1 or 2).

A paid to inform B (CASE 1 OR 2)

	A gets	B gets
Out is the final decision	5	5
In is the final decision, B chooses *Down*	9	10
In is the final decision, B chooses *Up*	1	14

A didn't pay to inform B (CASE 3)

	A gets	B gets
Out is the final decision	5	5
In is the final decision, B chooses *Down*	10	10
In is the final decision, B chooses *Up*	2	14

Instructions Part 2

We will now ask you to make guesses about the choices or beliefs of the participant you interacted with during part 1.

IF YOU ARE IN ROLE B:

We will ask you to answer two blocks of questions. One block will be randomly selected for payment (each block with equal chance).

In BLOCK B1, we will ask you to tell us for each of the possible values of p, **how likely you think A chose *In*.** Thus, you will answer 11 questions in total, one for each value of p between 0 and 100.

If this block is selected for payment, you will be paid according to your answer to the **question which corresponds to the realized value of p**. For instance, if the selected p was 80, you will be paid according to your answer to the question: "How likely do you think A chose *In* if her decision is implemented with a 80% chance?". You can earn 5 dollars for your answer to this question.

To determine your payment, we use a method under which you maximize your chances of winning 5 dollars by submitting what you truly believe are the chances that A chose In. More precisely, your compensation will be determined as follows:

1. You will first tell us how likely you think A chose In by picking a number among the options $\{0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$. For example, if you pick 60, this means that you think there is a 60% chance that A chose In.
2. Second, the computer will randomly draw a number between 0 and 100.
3. If the random number is equal or smaller than the number you submitted, then you will receive 5 dollars if A indeed chose In. For example, say that the random number is 24 and that you thought A was 60% likely to choose In. Because the number you submitted (60) is higher than the random number (24), you will receive 5 dollars if A indeed chose In and 0 if she chose Out.
4. If the random number is higher than the number you submitted, then you will receive 5 dollars with the probability determined by the random number. In other words, in that case, the chances that you receive 5 dollars are equal to the random number. For example, say that you thought A was 80% likely to choose In and the random number picked by the computer is 88; because the number you submitted (80) is lower than the random number (88), you will receive 5 dollars with a 88% chance.

What should you do when answering the question: "How likely do you think A chose In"? The best you can do is to choose the number that is closest to how likely you think A chose In. Indeed, suppose for example that you believe there is a 60% chance that A chose In but you choose to select 40. Then you will regret your choice if the computer for instance selects 50. Since the random number (50) is higher than the number you submitted (40), you will earn 5 dollars with a 50% chance. On the other hand, if you had declared 60, you would have earned 5 dollars in case A indeed chose In, which you think has a 60% chance to occur.

In BLOCK B2, you will answer two separate questions for each possible value of p (22 questions in total):

QUESTION 1: **Suppose A chose *In*.** We will ask you to tell us **how likely you think A paid to inform you that she chose *In*.**

QUESTION 2: **Suppose A chose *Out*.** We will ask you to tell us **how likely you think A paid to inform you that she chose *Out*.**

If this block is selected for payment, you will be paid according to your answer to the question which corresponds to **A's actual choice for the realized value of p**. So if A chose In (resp. Out) for the realized value of p, you will be paid for your guess in QUESTION 1 (resp. 2). Since you don't know A's choice or the realized value of p, you should report what you truly believe is the chance that A paid to inform you for each of the two cases (A chose In and A chose Out) and each value of p. In order to pay you, we will follow the exact same procedure as for BLOCK B1.

In summary, you will answer 2 blocks of questions, B1 and B2 (33 questions in total). You will be paid according to your answer for the realized value of p in one of the two blocks chosen at random. You will receive either 0 or 5 dollars for the randomly selected question.

IF YOU ARE IN ROLE A:

The B you were matched with was asked 2 blocks of questions for every value of p:

B1: How likely do you think A chose In?

B2 - QUESTION 1: Suppose that A chose In. How likely do you think A paid to inform you that she chose In?

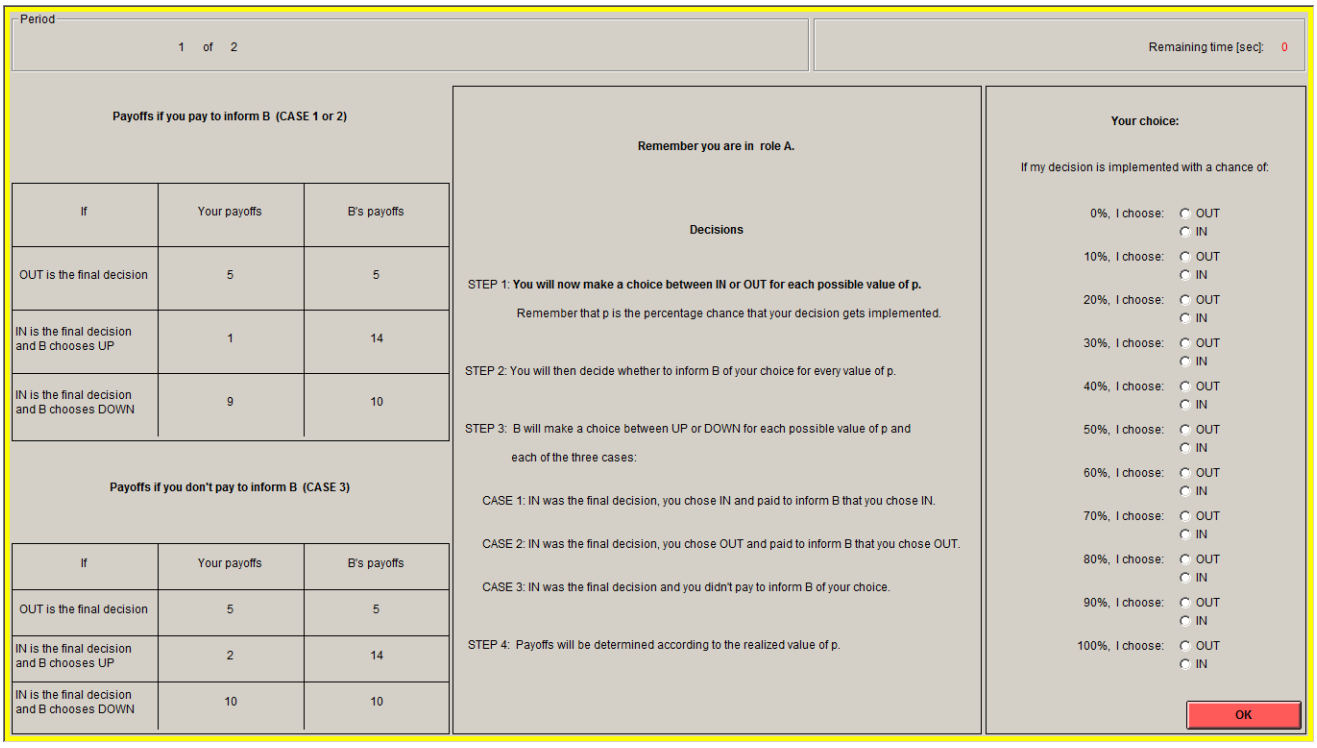
B2 - QUESTION 2: Suppose that A chose Out. How likely do you think A paid to inform you that she chose Out?

B was asked these questions for each possible value of p, making 33 answers in total. For each of these questions, B could choose a number among the options $\{0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$. For example, if B picked 30 in B1 when p is 40, this means that B thought you were 30% likely to choose In if there is a 40% chance that your decision gets implemented.

For each of the 33 questions, we ask you to **guess B's answer**. To determine your payment, we will select at random either B1, B2-QUESTION 1 or B2-QUESTION 2 and you will be paid 5 dollars if you guess correctly B's answer to the **question which corresponds to the realized value of p**. For instance, suppose that B1 is selected for payment and that the realized value of p is 40. If B picked 30 for that question, then you will receive 5 dollars if you choose 30 and nothing for any other number.

D Screen shots for Treatment *COM*

D.1 Action and information decision screens for A



Period

1 of 2

Remaining time [sec]: 0

Payoffs if you pay to inform B (CASE 1 or 2)

If	Your payoffs	B's payoffs
OUT is the final decision	5	5
IN is the final decision and B chooses UP	1	14
IN is the final decision and B chooses DOWN	9	10

Payoffs if you don't pay to inform B (CASE 3)

If	Your payoffs	B's payoffs
OUT is the final decision	5	5
IN is the final decision and B chooses UP	2	14
IN is the final decision and B chooses DOWN	10	10

Remember you are in role A.

Decisions

STEP 1: **You will now make a choice between IN or OUT for each possible value of p.**

Remember that p is the percentage chance that your decision gets implemented.

STEP 2: You will then decide whether to inform B of your choice for every value of p.

STEP 3: B will make a choice between UP or DOWN for each possible value of p and each of the three cases:

CASE 1: IN was the final decision, you chose IN and paid to inform B that you chose IN.

CASE 2: IN was the final decision, you chose OUT and paid to inform B that you chose OUT.

CASE 3: IN was the final decision and you didn't pay to inform B of your choice.

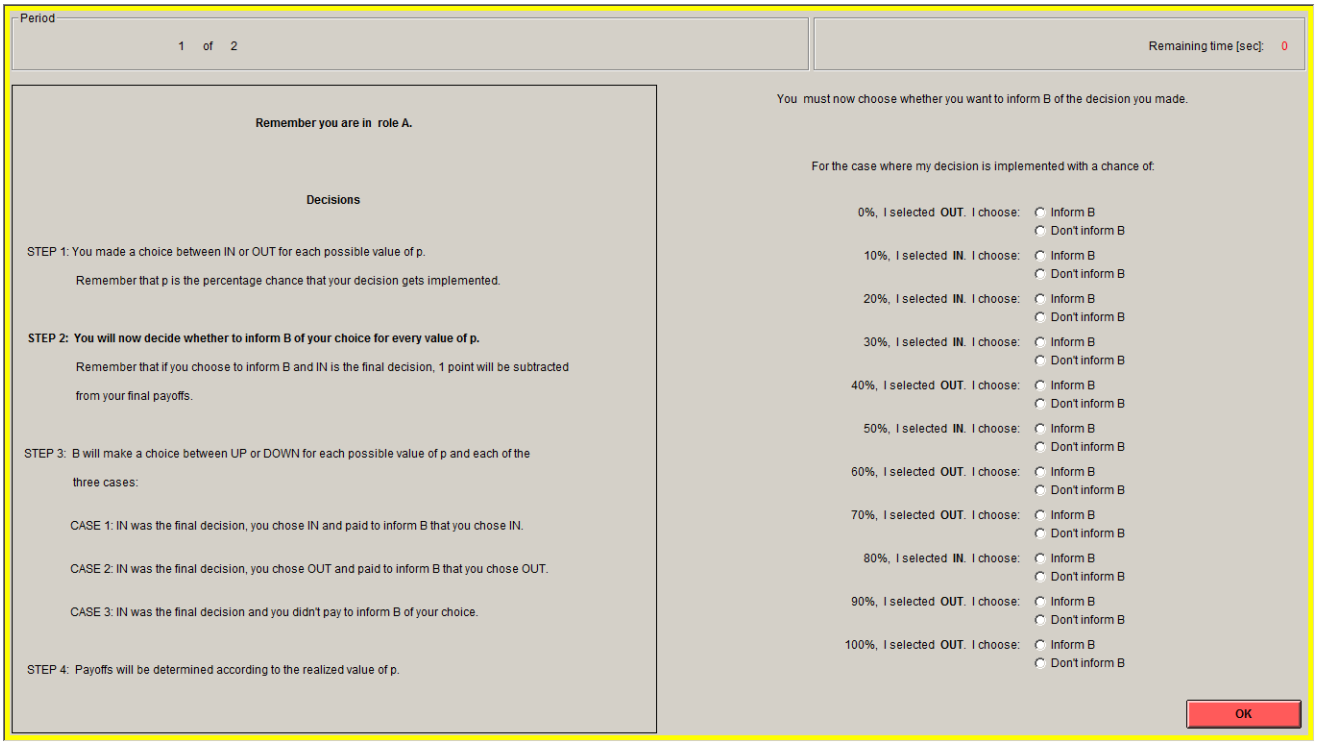
STEP 4: Payoffs will be determined according to the realized value of p.

Your choice:

If my decision is implemented with a chance of:

0%, I choose: OUT / IN
10%, I choose: OUT / IN
20%, I choose: OUT / IN
30%, I choose: OUT / IN
40%, I choose: OUT / IN
50%, I choose: OUT / IN
60%, I choose: OUT / IN
70%, I choose: OUT / IN
80%, I choose: OUT / IN
90%, I choose: OUT / IN
100%, I choose: OUT / IN

OK



Period

1 of 2

Remaining time [sec]: 0

Remember you are in role A.

Decisions

STEP 1: You made a choice between IN or OUT for each possible value of p.

Remember that p is the percentage chance that your decision gets implemented.

STEP 2: You will now decide whether to inform B of your choice for every value of p.

Remember that if you choose to inform B and IN is the final decision, 1 point will be subtracted from your final payoffs.

STEP 3: B will make a choice between UP or DOWN for each possible value of p and each of the three cases:

CASE 1: IN was the final decision, you chose IN and paid to inform B that you chose IN.

CASE 2: IN was the final decision, you chose OUT and paid to inform B that you chose OUT.

CASE 3: IN was the final decision and you didn't pay to inform B of your choice.

STEP 4: Payoffs will be determined according to the realized value of p.

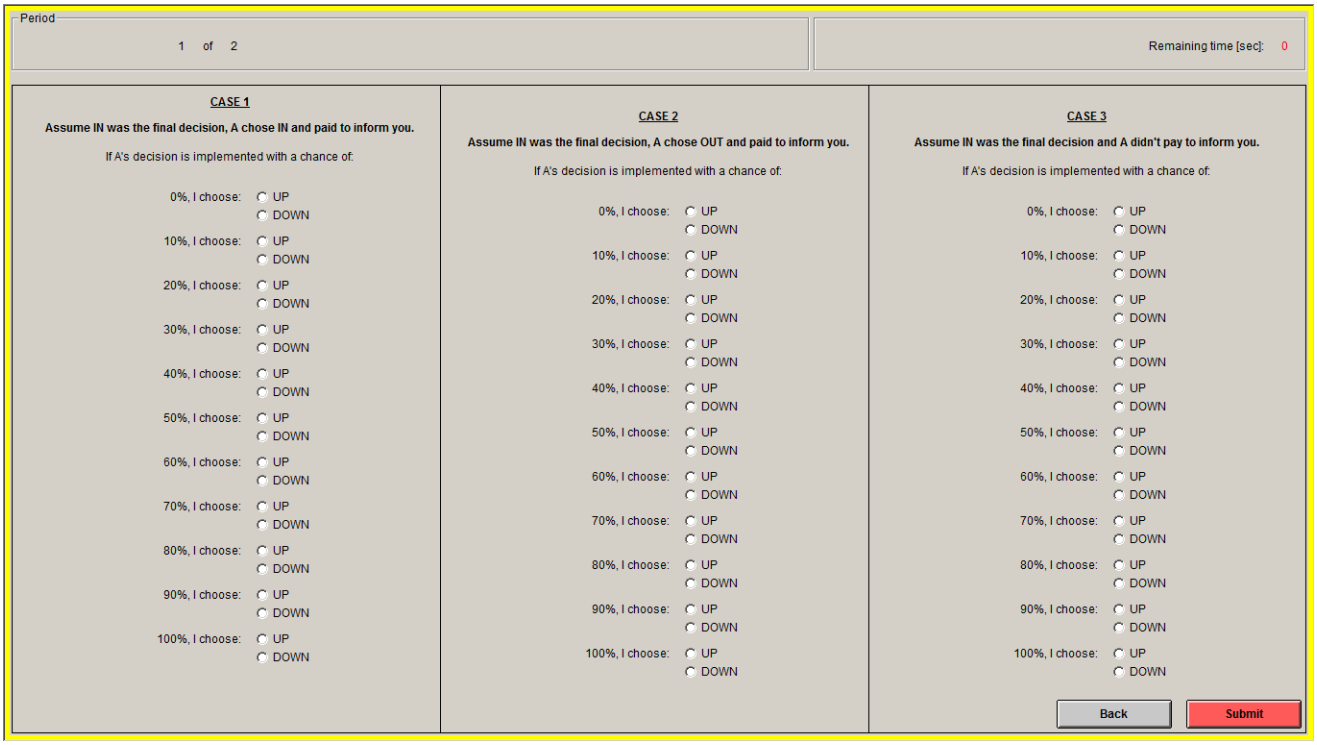
You must now choose whether you want to inform B of the decision you made.

For the case where my decision is implemented with a chance of:

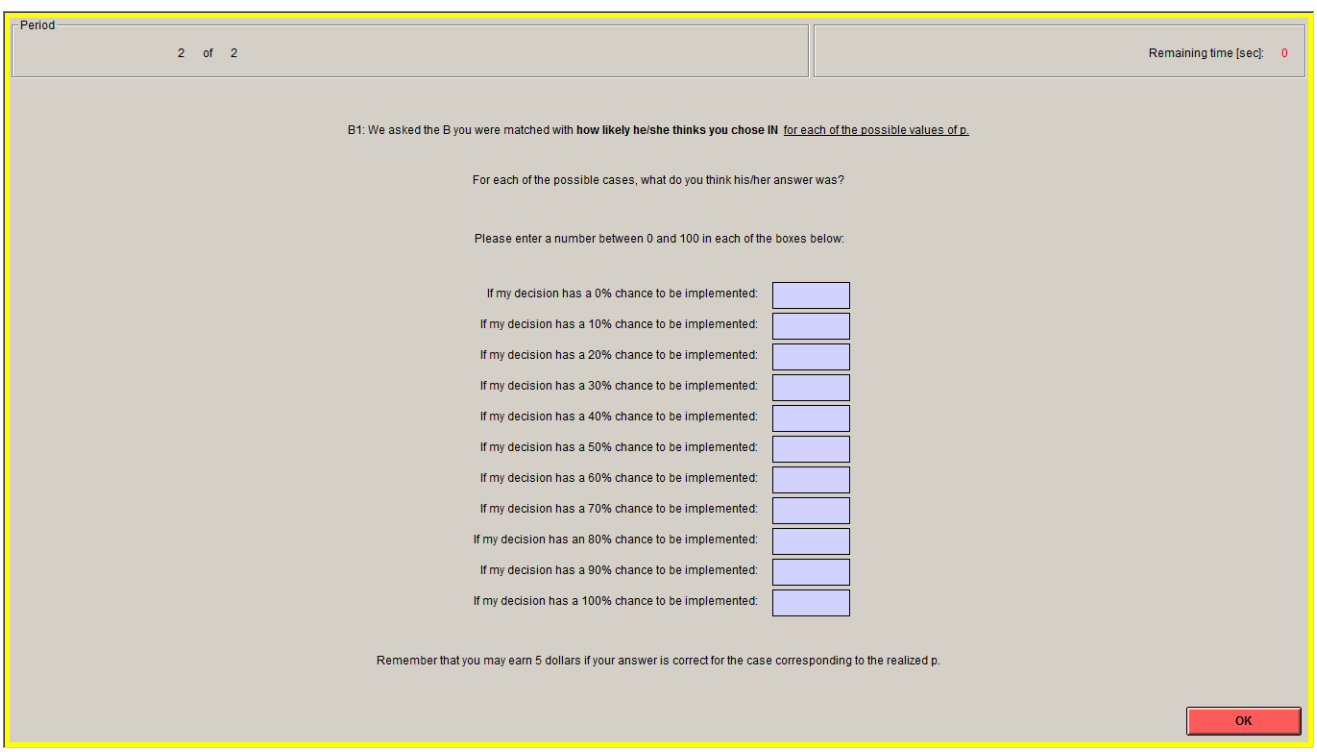
0%, I selected **OUT**. I choose: Inform B / Don't inform B
10%, I selected **IN**. I choose: Inform B / Don't inform B
20%, I selected **IN**. I choose: Inform B / Don't inform B
30%, I selected **IN**. I choose: Inform B / Don't inform B
40%, I selected **OUT**. I choose: Inform B / Don't inform B
50%, I selected **IN**. I choose: Inform B / Don't inform B
60%, I selected **OUT**. I choose: Inform B / Don't inform B
70%, I selected **OUT**. I choose: Inform B / Don't inform B
80%, I selected **IN**. I choose: Inform B / Don't inform B
90%, I selected **OUT**. I choose: Inform B / Don't inform B
100%, I selected **OUT**. I choose: Inform B / Don't inform B

OK

D.2 Action screen for B



D.3 Belief screens for A (B is similar)



Period

2 of 2

Remaining time [sec]: 0

B2 - QUESTION 1

For each value of p, we asked the B you were matched with the following question:

'Suppose that A chose IN, How likely do you think **A paid to inform you that he/she chose IN?"**

Please provide a guess of B's answer to each question by entering a number between 0 and 100 in the boxes below:

If my decision is implemented with a chance of:

0%, I guess:
10%, I guess:
20%, I guess:
30%, I guess:
40%, I guess:
50%, I guess:
60%, I guess:
70%, I guess:
80%, I guess:
90%, I guess:
100%, I guess:

B2 - QUESTION 2

For each value of p, we asked the B you were matched with the following question:

'Suppose that A chose OUT, How likely do you think **A paid to inform you that he/she chose OUT?"**

Please provide a guess of B's answer to each question by entering a number between 0 and 100 in the boxes below:

If my decision is implemented with a chance of:

0%, I guess:
10%, I guess:
20%, I guess:
30%, I guess:
40%, I guess:
50%, I guess:
60%, I guess:
70%, I guess:
80%, I guess:
90%, I guess:
100%, I guess:

Remember that you may earn 5 dollars if your answer is correct for the case corresponding to the realized p.

OK

E Exit questionnaire

Below is a list of selected comments made by subjects at the end of the experiment. In all treatments, subjects were asked to explain whether and why their decision depended on the value of p. In treatment COM, subjects in role A were also asked to explain whether and why they chose to communicate their action and subjects in role B were asked to explain whether their decision was affected by A's communication choice. The entire list of comments is available from the author upon request.

E.1 Comments of subjects in role A

- "Firstly, the role of A meant that I had the lower hand in this experiment. My decisions depended on the value of p. When p was equal to 50 or higher, I chose OUT in order to increase my chances of earning $5 over $2 (assuming that every individual is looking out for her/himself). As p lowered, I gave my partner the benefit of the doubt, hoping that he or she would choose DOWN." (ID 2, Session 3, NI)

- "When p is small, I take the risk of choosing IN, because I knew that B would most probably choose DOWN, since he/she has nothing to lose if he/she chooses DOWN, but I still wanted to take a chance and choose IN just in case the person wanted to choose UP. However, when p is big, I could not risk loosing money, so I had to choose OUT." (ID 12, Session 3, NI)

- "I thought that if my decision was less likely to be implemented, I would choose IN - I was in Role A and IN was a risky decision for me, so it would be more beneficial for me to just get 5 dollars than risk getting 2 dollars while trying to get 10. If, on the other hand, my decision was more likely to be implemented, I'd choose OUT since that is a relatively safe way to get 5 dollars. My earnings weren't as big as I expected though, so maybe that wasn't the best approach." (ID 8, Session 2, EI)

- "I was side A so I had a slight disadvantage compared to side B. I choose a risk averse method with most of my choices resulting in OUT. The higher the p, the more likely I was to choose OUT. At lower values of p, my choice was less likely to be chosen so I went with IN." (ID 16, Session 4, EI)

- "I informed my partner of my decision for those cases that I chose IN. I felt this would encourage my partner to show generosity rather than act selfishly by showing them that I had appealed to their good nature instead of making the more cynical decision. However, in cases where I chose OUT, I saw no advantage to informing my partner of the decision, as I was basically making the assumption by choosing OUT that they would act selfishly." (ID 4, Session 2, COM)

- "I made my decisions IN/OUT according to the value of p in that the less likely I was able to implement my decisions, I was more inclined to choose OUT, however if there was a substantial amount of chance in my decision being implemented, I automatically chose IN, because I would count on the hope that I could have a higher payoff. I didn't inform B of my choice until the p was high enough for me to think that I had sufficient control in my decisions being implemented, because I found that that it might be a waste of my payoff earning." (ID 6, Session 3, *COM*)

- "I chose to inform B when I chose IN because I wanted to look like I was willing to help them out, I guess? But when I chose OUT I didn't tell them because it would always result in them earning less money. So I think I wanted them to be motivated to press DOWN in cases where they knew I chose IN. Not sure if this is logical or not but it's how I was thinking." (ID 5, Session 4, *COM*)

- "I did not pay to inform B of my choice in any case (unless my memory fails me) because the system did not actually take this into account in real time. If this game were strategy based, where player B knew my decicions before making their own, I might have informed them, but since my choices had no impact on theirs it did not matter to me and I did not want to lose $1 of my potential earnings." (ID 6, Session 4, *COM*)

- "I sometimes chose to inform B when I chose IN to maybe sway him not to choose the payoff best for him in hopes we could both come out with almost equal payoffs." (ID 17, Session 5, *COM*)

E.2 Comments of subjects in role B

- "The higher the value of p, the more likely i was to put up because a would be reluctant to choose in." (ID 14, Session 1, *NI*)

- "If A's choice is implemented and A chose in, I would appreciate A's decision and choose down. If p is higher than 80, which means A's choice is more likely to be implemented, i choose down." (ID 21, Session 1, *NI*)

- "I was B so I tried to guess what they liklihood I thought A was to choose In or Out, and then in an effort to maximize both earnings (trying not to short-change A) I picked DOWN for all p values but 0, 10, and 20." (ID 13, Session 2, *NI*)

- "The higher the probablilty the more frequently I decided to go with Up because it would definit me more than choosing Down would. When p was very low, I put down and as p got larger, I chose up instead." (ID 10, Session 3, *NI*)

- "My choices for up/down did depend on the value of p. If the value of p was low, I was more inclined to choose the up because I would receive a higher payoff. If the value of p was high, I chose down because the decision was greatly impacted by A. Thus, A would not want a higher payoff for me, so I chose the decision with equal payoff for the both of us." (ID 9, Session 4, NI)

- "My decisions depended on the value of p in that if A chose out and his decision defninitely was going to be implemented, I was more likely to choose down because my decision would not have mattered. The higher the probability that in would be the result, the more certain I was to choose up." (ID 14, Session 1, EI)

- "I tried to cover myself, in case ROLE A proved to be more selfish than I anticipated, and chose UP/DOWN accordingly with what I felt ROLE A deserved. So if ROLE A chose to be IN and her decision would be implemented with a chance of 100%, it seemed to me that I should choose DOWN as they were cooperative, and thus in my opinion, more deserving of their share. If I thought they would choose OUT then I would pick UP, in order to sway the profit my way." (ID 19, Session 3, EI)

- "My decisions were influenced by p only for when p=100 and p=90 and wanted to reward the corresponding A. All other decisions were made out of self-interest." (ID 8, Session 4, EI)

- "I made my decisions between up and down depending primarily on the value of p and cases. In case 1 I put down for half and up for the other half. I figured that the higher the chance of A's choice being implemented the more likely they were to choose in and simultaneously I wanted my decision to be mutually beneficial so I chose down so in that case, if it was chosen, we would both recieve ten dollars. However, for lower p values, where it depended less on A's choice, I did not feel that obligation and chose up to increase my own winnings." (ID 13, Session 4, EI)

- "It depended on how sure I was of A's choice. If I knew A was going to choose out, my decision of up/down did not matter. If I knew A was going to choose in, my decision depended on the value of p (how sure I was that the final decision was going to be \"in\") and hence, how likely my decision was going to matter. I chose to put down for some of the higher options because I felt uncomfortable with the fact that I would be earning more money at the expense of someone else, however the temptation of earning that extra money 'for sure' at 100% in was too great." (ID 20, Session 2, COM)

- "I made my decisions on the likelihood of getting a higher payout. As B, I chose up or down based on p and the probability of my decision mattering (ie. if A chose in or out, if A chose out and the probability of it being out was high, then my decision did not matter as much

but I chose up or down still based on whichever would give me a higher payout if the decision was in)."

- "I played differently based on whether A paid to inform me because if A informed me that they chose out, then we both make the same amount in the end. If A did not pay to inform me, then I assume A chose out or there was high probability of the final decision being out or A chose in and did not want to risk me choosing up (vs. down where they would not have made as much payout)" (ID 4, Session 5, COM)